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Using Technology for Real-Time Coordination of Work

A study of work and artifact use in the everyday activities at SOS Alarm

Maria Normark



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Department of Numerical Analysis and Computer Science

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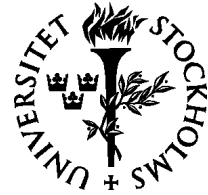
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Abstract

Work in control rooms, or so-called Centers of coordination, places demands both on humans and technology. The people working there have to be able to make quick decisions as well as be alert during less busy times. The work has to be coordinated within the group, since the operators are much depending on each other's work. This places special demands on the technology; it should be fast, trustworthy and easy to manipulate so that the complexity of the work is reduced. This type of work has been a source of interest in the research area of Computer Supported Cooperative Work, especially so since the needs for real-time coordination is great in these settings.

SOS Alarm is a company that is responsible for managing the telephone calls made to the emergency number 112 in Sweden. They have 20 centers covering Sweden. This thesis present an overview of the SOS operators work; how they coordinate the information and tasks between them; how the technology supports that work. This study presents a fully computerized setting, compared to many other studies of centers of coordination that are not.

This ethnographic workplace study has been inspired by ethnomethodological perspectives as well as earlier field studies of work and technology use. Its main results are a description of the work practice and technology at two centers, implications for design of a new computer aided dispatch system that is currently developed at SOS Alarm and general design ideas for control rooms.

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1. Introduction

- SOS one one two, what has happened?

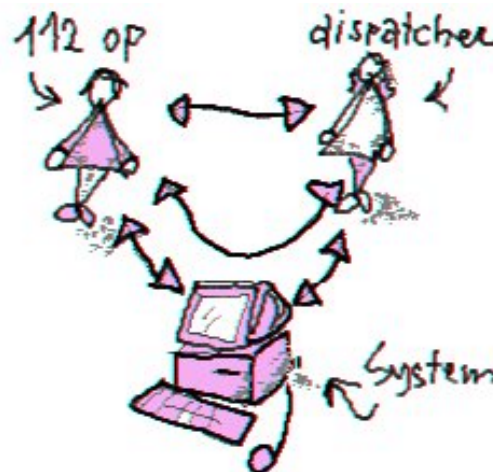
- Well, I'm standing here outside the grocery store and a cyclist has hit an old lady and she cannot rise from the ground. Her dog went nuts and bit a by-passer and there is a truck here that was loading groceries it can't get out and it's blocking the traffic.

The variations of incidents that are called in to the emergency call centers that receive the 112-calls in Sweden (SOS Alarm) are vast. The SOS call-takers', or, as they are called, operators', work is to try to make sense of the call, collect information, assess what is said, make a decision, send out rescue units and monitor the development until the situation is safe. This work involves contacts with several groups, citizens, ambulance drivers, rescue service, police, etc. There are 20 centers covering Sweden that each work separately.

The study presented here was made against the background of a planned new computer aided dispatch system that will enable redirection of emergency calls to the least busy center in Sweden. This means that coordination amongst operators is likely to be needed across geographical distances, and raises several questions. What kind of coordination is done today? What is the role of technology in that coordination? What does geographical proximity mean both between the operators at the center but also in relation to the area they are responsible for? Initially, SOS Alarm asked us to look into these questions (reported in Helgeson, Lundberg, Normark, Pettersson, & Crabtree (2000)). The work practice and technology lab, led by Bo Helgeson, professor in Human Work science at the Blekinge Institute of Technology were asked to do research in order to do a study of work and technology use amongst the operators. I had the opportunity to be a part of this project, which suited my interest in coordination and technology well.

In this study, the focus is on the issue of the work of the operators, and the coordination between them. What, for example, is coordinated through the computer system and why? What is not?

The focus will be put on three different parts: the receiver, the dispatcher and the computer system. These three parts could be viewed upon in terms of individual and collaborative work. Some of the operators' work is not relevant to others, some is and some is done in collaboration with another operator. What I am interested in can (somewhat



crudely) be pictured like this:

Figure 1: The coordination possibilities inside and outside the CoordCom system

The cases are almost always communicated between the operator that received the call (the 112 operator) and the dispatching operator. The part of the work I focus on is from the incoming call until the dispatch is made, or the call is ended. Drawing on the analysis made by Artman & Waern (1999) where the initiation of a SOS case was seen as a shared task between the 112 operator and the dispatcher, the analysis here adds the system as a part of the contributor to the tasks' fulfillment

This workplace study aligns to the Computer Supported Cooperative Work research tradition where control rooms, sometimes referred to as Centers of Coordination, has been a source of interest since the mid-eighties, see Bowers, Button, & Sharrock (1995); Heath & Luff (1992); Hughes, Shapiro, Sharrock, & Anderson (1988); Suchman (1993b) to name just a few. My interest in control rooms began with the air traffic control study I participated in earlier, see e.g. Berndtsson & Normark (2000). I find the varying work load and the complexity, typical for this kind of work, useful for learning more about the use of technology and tools for coordination.

Defining coordinative work

An assumption is that a work setting is an arena where the actors to a large extent depend on tools as coordinative mediators (i.e. notes, PM's, documentation, orders, forms), and that the

common goal makes it necessary to coordinate. In the thesis I will discuss how social mechanisms are inscribed in different artifacts and how a competent member interprets different objects as resources for social interaction. I make a distinction between:

- cooperation: working for a common benefit
- collaboration: working jointly
- coordination: working interdependently, adjusting to the immediate work of others

The perspective is thus on individuals rather than groups or organizations. Earlier, the question of whether all work is social or not was mentioned. My assumption is that most if not all human conduct is socially oriented. However, not all work is coordinative. In the following, Schmidt discusses the balance between individual and cooperative work in the sense that I would describe coordinative work:

"While work is always socially situated and socially organized, the very process is not always cooperative in the sense that it requires and involves multiple actors who are thus interdependent in their work." (Schmidt, 1994)

Research interest and background

A fundamental part of any social activity is that we orient ourselves in a social way. Simple examples of how we physically orient ourselves is the way we avoid bumping into each other at the pavement or the way we orient ourselves to show that we are queuing. We also use language and signs to socially inform others of our intentions and opinions, etc. The way we choose to orient, adjust, express etc ourselves is largely dependent on our culture. Greetings, for example, differ between countries, cultures and groups, some do nose rubbing, some shake hands, others give a high five and so on. The important issue is that we make our expressions of social interaction intelligible to others in a way that is dependent on our common cultural knowledge. These cues often involve artifacts. One assumption in this research is that professional settings have their own cues and signs that symbolize certain important information for the group. What does for example certain information mean and why is it important? Earlier, machines were generally large, noisy, heavy and easier to read off the status, consider for example the typewriter. One usually did not bring it around; it stood where it stood and one knew what it was used for and it was easily heard through doors. These features could be used as cues by others to understand what was happening. Today, when we use computers for various kinds of tasks it is much more complex to understand from the setting what is happening and what a tool is used for. It is also more difficult to manipulate the computerized material in a way so that others understand what one wants to highlight or put emphasis on. One cannot make dog ear foldings or underline, for example, on a computer screen as easily and as obviously as with paper. And if one does, it is difficult to be certain that ones collaboration partner knows that feature in the program and will notice it. It is thus important to study how people in complex work using computers, deal with them to achieve interaction.

I am interested in the cues that are needed at a SOS center in order for an operator to be able to handle a case, to make sense of it, and how it is presented to them. How are artifacts in general and the computer system in particular part of the sense-making work?

In order to learn more about what role the computers play for interaction in the work of emergency operators I will focus on the following:

How is real-time (as-it-comes) coordination done in emergency dispatch and how are the features of the computer system vs. other tools a part of it?

CSCW research

This research is mainly performed in the Computer Supported Cooperative Work tradition. I draw upon the work of Kjeld Schmidt who writes:

”/.../cooperative work arrangements arise from and dissolve into individual work. More than that, the boundary between the individual and cooperative work is dynamic in the sense that people enter into cooperative work arrangements and leave them according to the requirements of the current situation and the technical and human resources at hand.” (Schmidt, 1993).

Researchers in the Computer Supported Cooperative Work (CSCW) tradition have rather different interests and backgrounds. One background might be an interest in technology that enables coordination, a more technologically oriented perspective. These researchers take interest in network protocols, algorithms, mobile technology and other solutions that enable people to share data. Another background is an interest in design for groups. More and more the concept of one computer - one user is replaced with a perspective of human to human - computer interaction. Typical questions for researchers from this perspective is evaluation of different systems, for example chats, instant messaging, haptic interfaces and so on. A third kind of research interest in CSCW is of the complex skills that professionals have and how they deal with tools and technology for coordinative purposes. Often, this interest produces studies of the current situation at different organizations and some implications for a new design. These studies are usually referred to as workplace studies. This research is perhaps mainly in line with the last path.

Of course, Computer Supported Cooperative Work is a research topic that has followed in the trace of the computer use development. From the beginning with the main idea to enhance groupware it has developed into an area more emphasizing group activities.

The "Computer Supported" part of the research field can be understood in much wider terms and does not necessarily have to involve what we traditionally think of as computers. The use of other tools may as well be helpful in informing us how cooperation mediated by tools and what designs that would be useful. Another assumption is that we do not know so much how tool-mediated cooperative work is carried out. The Cooperative Work part of CSCW has continuously been influenced by certain sociological theories. One thing that is considered important by many CSCW researchers is to study cooperative work in its natural setting, at the place that they occur. Bannon and Schmidt write that CSCW is:

“...an endeavor to understand the nature and characteristics of cooperative work with the objective of designing adequate computer-based technologies.” (Bannon & Schmidt, 1989)

A general problem of HCI is to make design more use oriented. A conclusion is that many new programs and solutions fail because of their unsuitable design for the users. If one agrees that work is essentially social, it becomes obvious that we must learn more about how technology and tools are used in general and for human to human interaction in particular. There are many rational descriptions of work that often serve as a foundation for design. However, as many studies have shown, the formal description often lacks an important understanding of the work that is going on. How can we learn more about how technology is used in everyday work? This is where the workplace studies come in.

Reader's guide

After this introduction, a chapter on workplace studies follows. It contains a description of my method; ethnography. It is a qualitative method where one does studies at the field of interest, often over an extended period of time. Further it discusses the theoretical influences that are often connected with workplace studies. The next chapter, Concepts of work and technology, contains results from other studies of work and technology use. The theoretical discussion is ended by a summary of the theoretical discussion.

Thereafter, the field study of emergency call receiving and dispatch at SOS Alarm is introduced (The SOS Alarm study chapter). It is followed by an overview of some of the domain-related research. The field studies begin in chapter 7; Work description and coordinative concerns. Thereafter four authentic cases are presented and discussed in chapter 8. The cases are followed by a chapter about design considerations. The chapter Concluding discussion and future work ends the thesis.

2. Workplace studies

For some decades now, studies that combine an interest in technology use, work and design have been done within the CSCW area. These are usually referred to as workplace studies. They usually include long-term field studies and attention to the work that is carried out in the setting in order to get detailed accounts the situated work and technology use. According to Plowman, Rogers, & Ramage (1995):

"Their /the workplace studies/ primary role is to convey the importance of the sociality of work by shedding light on the complex actions and interactions that occur." (p 309).

Why do we need workplace studies to understand work and technology use? Schmidt (2000) explains the problem of designing for cooperative work:

"We are all experts /on cooperative activities/ from our everyday experience. And yet this quotidian insight can be utterly misleading when applied to the design of systems to support cooperative work."

As was mentioned before, people in certain professions, in the specific cultures at work read information differently and act upon cues that the layman may probably not understand. Even if we have the same general culture, where a nod means yes for example, it may mean a lot more in a specific setting. The problem of designing for abstract, intellectual work is that the purpose is not obvious. If someone considers designing a doorknob, s/he has some experience of using a doorknob. If someone is designing a computer aided dispatch system for SOS Alarm, it is likely that the designer has no experience of this and does not know the possibilities and constraints of the work.

Schmidt further suggests (Ibid.):

"The primary role of workplace studies in CSCW is thus to dismantle the common-sense conceptions of cooperative work, take them apart, unpack and disclose the hidden practices of articulation work, and thus give us access - analytically and conceptually - to the intricate ways and means of the production of social order in cooperative activities." (p 4)

Workplace studies are often related to ethnography and ethnomethodology (two subjects I will get back to later on), but are also carried out by researchers with other perspectives and analysis methods, e.g. Distributed Cognition (Halverson, Harwood, TJ, & CR, 1992; Hutchins, 1995) and Activity Theory (Bardram, 1997; Engeström & Middleton, 1996). The main reason for doing workplace studies is that there is no other way to study coordination in the wild (to paraphrase Ed Hutchins), how it occurs when the motivations, reasons and conditions for doing a task are authentic. I will get back to the reasons for doing workplace studies in the method chapters.

There has been a debate about the role of workplace studies. Mainly, the discussion has concerned the relationship between these studies and design. How exactly can workplace studies inform design? Several suggestions have been made on how this can be done, see e.g. Karasti (1997); Randall, Hughes, & Shapiro (1994). The question about what will happen to the workplace studies in the area of CSCW is at the moment rather unknown. The problem lies in how to use ethnography to inform the design in an effective way. One way to use the ethnographic studies may be to sum up what has been learned as suggested by Bannon (2000). There are several issues that are reoccurring in these studies that might serve as foundation for a general description of coordinative work and technology, I will get back to some suggestions in chapter 3.

Ethnography

In CSCW the ethnographic method is mainly used for the focused purpose of doing studies of work. The method was adopted from sociology and anthropology, where it from the beginning was used to study non-western cultures. By looking at complex work situations (e.g. control room work), ethnographers in CSCW try to understand how collaboration is performed *in situ* and how different artifacts in the environments affect and enable coordination. Two main reasons for using ethnography in these areas are to survey the conditions of a specific kind of work and to document the skills and practices of the people working in a certain profession. Another reason is to capture everyday, taken-for-granted cultural assumptions in which activities take place. As the word 'ethnography' implies¹; the outcome is an extensive written text. Analyzing the work from the standpoint of coordination is usually a part of the research.

By spending time in the workplace, ethnographers try to understand the skills of a certain profession by interacting with the professionals and studying as well as trying to

¹ **Ethno**: people, cultural group; **-graphy**: writing

observe and uncover the hidden practices of the work. A special interest is often the discrepancy between how people think that they do things and how they really do. Research has shown that it is difficult for us to estimate time and remember in detail how we do things. Nyce and Bader give the following example

"To give an example of how practice and thought disjoin in everyday life, let's look for a moment at eating and dieting. It is quite common for example to ask people who wish to lose weight to keep a "food diary". The purpose of this diary to help people become aware of the routine, taken-for-granted aspects of their life like how, when, where, and why we eat the way we do. One would certainly think that a person would be aware of what they are doing while they are eating. Often we are not."(Nyce & Bader, In press (2001))

A belief is that many of these misconceptions of what and how we really do things are reasons for computer systems failing to be useful. A good ethnography can thus be used for identifying the correct problem that needs to be solved in a new design. Ethnography is one way of capture the *situatedness* of specific skills. This knowledge is very useful when thinking about a future support for an activity.

In order to understand and be able to analyze the skills and practices in connection to a certain setting, the ethnographer collects descriptions, impressions, stories, pictures, cases, history, drawings, and all the information that one can get in order to understand how the practitioner work. It is easy to drown in this material. It is also easy to loose the focus of the investigations. Ethnographers are often criticized for their fascination and loyalty to their subjects and for the tendency to pay tribute to the clever ways that activities are performed. In studies aiming at to contribute to developments of new products, this has especially been an issue, since "cheering" at the ways things currently are done leads nowhere in designing. However, I believe the problem of aligning ethnography and design is one of different perspectives, more than ethnographers' possible lack of interest for future design. Ethnographers view the setting in a holistic, interdependent and locally situated perspective, while designers divide the problems into small, logic pieces.

I believe that bringing in knowledge of how the work is carried out and thought about in the context will add a new and important aspect in design projects. Nyce and Bader (ibid.):

"Individuals /experts/ are very often unaware of their own basic cultural assumptions or the relations of power and authority in which they live an around which they create their lives."

This is suggesting that a person that comes from the outside and is unfamiliar with the setting may collect a lot of contextually important issues that are difficult to abstract by an expert from the within.

Nyce and Bader (Ibid.):

"This raises of course the more general issue of whether rule bound description can be adequate at all when it comes to human action."

However a bold statement, which has been extensively debated in the CSCW field (see Schmidt, 1997; Suchman, 1987), I do believe that the actual work and cultural setting needs to be considered during the design process.

Sometimes design work and its process are compared with architectural work. I think that much of the processes and practices used by architects would be useful, for example studying successful design, having a well-defined idea of how the product; the building is supposed to be used and so on. However, there is a large difference between designing a physical space from a virtual one. Understanding how people practically would like to use a building is easier than to understand how people intellectually would like to use a computer system for a specialized purpose. When designing a house one also has the advantage of knowing the physical restrictions of both the material and the body and the general cultural language. One can for example imagine pushing a hospital bed through a doorway and understand that it needs extra space. It is much more difficult to understand what, say a SOS operator, intellectually needs to coordinate to a colleague in order to hand over a case.

There are many complexities concerning the design process of new computer systems, one being the relatively new tradition (compared with e.g. architecture), so I think that it will develop much. One step I would like to see is a further use of ethnography and workplace studies, either in the form of detailed analysis of a specific setting or the experiences from a set of ethnographies. I will get back to the issue of using ethnographies for design later on.

Ethnomethodology

The use of ethnography as a method often has a more or less obvious relationship with ethnomethodology in CSCW, basically because of the interest in how things are carried out in everyday life. It is a theory in that sense that it attempts to describe how social life is organized. However, since the interest is of life *in practice* and not in theory, ethnomethodologists usually do not call their perspective a theory. Ethnomethodology was coined when Harold Garfinkel wrote his book *Studies in Ethnomethodology* 1967 (Garfinkel, 1967). In general it is assumed that social order is always there but takes different forms in different settings. The main question then in ethnomethodology is; how is it possible for us to make ourselves understood? What methods or cues do we use to achieve social interaction? When do we fail to achieve understanding? Alain Coulon describes ethnomethodology:

“The scientific project of ethnomethodology is to analyze the methods, or the procedures, that people use for conducting the different affairs that they accomplish in their daily lives./.../The “methodology” is used by the members of a society or of a social group in a banal but ingenious way to live together; these ethnomethods constitute the corpus of ethnomethodological research.” (Coulon, 1995)

Ethnomethodology focuses on what is important for the continuous construction of social interaction in a given setting. Alfred Schutz’s work about Phenomenology was an important

influence for the development of ethnomethodology. Schutz was interested in how people construct common understanding in activities where the members may have different observational perspectives, different goals, motivations or intentions. An example taken from Coulon's book is the activity of going to a football game. Even if we agree on the different qualities in experiencing the game through paying more or less money depending on where we want to sit (or stand), we still agree on everyone seeing the same game, and thus have an intersubjective knowledge of the game. Schutz meant that this knowledge derived from two 'idealizations'. One is that we are able to imagine things from different perspectives (not only visually) and the other one is that the spectators presumably have similar prerequisites, we assume that people go to a football game by similar intentions, in spite of individual differences. These two idealizations are the foundation for "the general thesis of reciprocity of perspectives" (Coulon, 1995).

In ethnomethodology it is assumed that social relations are constructed and interpreted in the same moment as they occur. Since we are continuously interpreting our environment, it is interesting to try to understand how humans make sense of the context we are in. Coulon writes:

"The relation between actor and situation is not stable and unchanging, produced by cultural contents or rules; it is produced by processes of interpretation." (Coulon, 1995)

By being interested in the situated activities constructing the social reality, ethnomethodologists often focus on language and how the conversation (both verbal and through body language) contributes to the intersubjective knowledge and understanding. See e.g. Heath and Luff's work on visual conduct (Heath & Luff, 1991b) or conversation analysis, e.g. Bowers, Pycock, & O'Brien (1996).

The ethnomethodological perspective tries to shed some light on social order, how people act to arrange an understanding, thus this is an important part of the perspective. Whalen (1991) writes:

"...how members of society /.../ achieved some interactional effect."

This is the core of ethnomethodology, trying to understand how interaction is possible and how we make ourselves understood. What methods do we use to give others cues about our intentions and opinions, etc.? Garfinkel put forward the question of "What to do next?" as an area of interest in social interaction. What "signs" do we act upon that trigger us to act?

Instead of trying to abstract mechanisms of social behavior, ethnomethodologists' focus on situated practice. The expression 'community of practice' (Lave, 1991) is used as an assembly term for the specific methods, practices, and skills that are used in a specific community (e.g. amongst certain professionals). It can, of course, be considered both at a low level, as certain professionals and at a high level, for example people living in the western part of the world. The competent member is a person who is recognized by others in a community as such. In strict ethnomethodological sense a competent member is someone who can pass as a member of the community. Thus, it has less to do with the actual competence and more with the capability to perform as a member. At SOS alarm, the focus of my studies, it would be quite problematic to falsely pretend to be a competent member, so

for the sake of my further argumentation I assume that a competent member is what it looks like. The focus on competent member, however, is to find out the certain meaning that is derived from a certain act, from within, and is unrecognizable to laypersons. A nod, for example, may mean "hello", or "go ahead", or as seen amongst air traffic controllers; "I have recognized that particular complexity and I'm taking care of it". This meaning, greatly connected to the context, is of interest to ethnomethodologists. How does a competent member make sense of the signals, cues, and signs that are available in the setting? What are of importance, and what is filtered away? Suchman (1995) writes:

"Making sense and use of representations of some aspects of the social world involves our own positioning in relation to what we are seeing as much as any meaning inherent in the images themselves." (p 63)

Ethnomethodologists focus on the common-sense experience of interaction.

Garfinkel writes:

"Ethnomethodological studies analyze everyday activities as members' methods for making those same activities visibly-rational-and-reportable-for-all-practical-purposes, i.e., 'accountable' as organizations of common place everyday activities." (Garfinkel, 1967)

Every-day tasks are performed by a qualified member in a certain community of practice (and are thus rational for the other members). We are making accounts to explain and help others maintain order through implicit or explicit explanations. Coloun writes

"To say that the social world is accountable means that it is describable, intelligible, reportable, and analyzable. All of these features are revealed in the practical actions of the people. The world is not given once and for all, it is constituted in our practical accomplishments. " (Coulon, 1995, p. 25)

This issue is interesting for the sake of design. How can a tool be designed to support accountability of social actions?

Situated action/situated practice

That action and practice are to be understood as situated within the context as they are taking place is a common assumption in CSCW and is very much a part of the ethnographic/ethnomethodologic approach. The concept entered the CSCW community at large with the arrival of Lucy Suchman's influential book (Suchman, 1987). This book has led to a debate about the relationship between pre-made plans, instructions, and the actual carrying out of an activity. It can be illustrated by the following simple example. Consider that you would make a rather detailed plan before you would do an action, say for example driving downtown. Your plan is first interrupted by a large lorry parked outside on the street so you have to take the other way. Your car stops and you cannot get it going, so you take the bus instead. I want to show that humans are flexible and can think of several ways to do the same thing, it is the goal and purpose that is important and not the way of getting there. If comparing this way of accomplishing activities to those of machines, there is a fundamental difference. Machines, or computers, are mostly constructed to do things in a

certain way and they do not feel motivation for the outcome. The point, as I interpret it, is that if tools are constructed to demand a strict, detailed plan of how an activity is to be performed, then there will always be interruptions because of the different approaches of machines and humans. The critique that has been made to Suchman's arguments is basically that in many settings, plans are the only way to coordinate and arrange work so as to keep the complexity down. An example of a "needed plan" is the checklist that the pilots use before takeoff (Schmidt, 1997). The usefulness of plans are difficult to discuss because it is impossible to decide at what level of detail a plan is acceptable as a guideline for conduct (and Suchman does not discard plans as such). This debate and research are of extra importance when studying design and tools for complex settings as control rooms, because of the professionals' need to be able to predict each others' actions, but their need to be able to create unconventional ways of solving problems that occur, perhaps never used before. Thus the technology needs to stick to an order of how things usually are done, but at the same time, the order (plan) must be alterable to adjust to new situations.

Working division of labor

Knowledge and action cannot be placed but are produced locally, situatedly within a context. It is a continuous work on sharing and dividing labor. It means that any attempt to create a formal chart of work will reflect only one picture of the multitude of possibilities in which that work is done. When describing work as of the product or service that moves around in an organization, its production is disregarded and the complex coordination work that made it move on are still clouded. If the working division of labor is not described, new design may hinder or destroy the common performance of work. The term egological is used to describe the practical, ongoing, division of labor. Calvey, Hughes, O'Brien, Rodden, & Rouncefield (1997) write:

"A small set of viewpoints emerged; - the setting of the work (ecology), the social context of the work (flow of work) and the practical organisation of the work (egological) taking place; - each presenting a particular focus on the social organisation of work activities and chosen in order to highlight relevant aspects of the sociality of work. " (section 5)

Anderson, Hughes, & Sharrock (1989) write:

"it is possible to conceive and investigate the division of labor as an egological organization of thematic relevances and structures of knowledge" (p. 170)

The main point is that people adjust to the situation at hand. If the secretary is busy, someone else will answer the telephone, even if it is not part of his or her formal job description. If the schedule tells someone to do a print job and the printer is broken, s/he can adjust to the situation and help somewhere else where it is needed. This continuous adjustment to the situation at hand is important when designing for coordination. The flexible adjustment to situated action compared with the rigidity of a formal chart is the reason for studying the setting from the inside.

Centers of coordination

The term centers of coordination was first used by Suchman to describe the characteristic work that is going on in command and control rooms and similar settings. Centers of coordination are settings where collaboration is crucial and updates of the current state of work need to be made in real-time. Many studies have reported on the awareness of 'what is going on in the room' as an important and crucial factor (Bowers & Martin, 1999; Heath & Luff, 1991a; Norman, 1993; Suchman, 1996). Typical centers of coordination are nuclear power plant control, air traffic control, and train control, but can also include less life critical control work, for example luggage handling at an airport. Suchman (1993a) writes

"Centers of coordination are characterizable in terms of participant's ongoing orientation to problems of space and time, involving the deployment of people and equipment across distances, according to a canonical timetable or the emergent requirements of rapid response to a time-critical situation." (p. 42)

Often the subject of control, for example airplanes or trains, is not within reach for the controllers or operators. This means that they are heavily dependent on representations and tools to read and manage the current state of affairs. Further (Suchman, 1993b) writes:

"Centres of coordination are designed to maintain two contradictory states of affairs. On the one hand, to function as centres requires that they occupy a stable site to which participants distributed in space can orientate, and which at any given moment they know how to find. At the same time, to coordinate a system of widely distributed activities, personnel within the site must somehow have access to the situation of others distant in space and time. A job of technologies in such settings is to resolve this contradiction through the reconfiguration of relevant spatial and temporal relations."

From the two studies I have done of centers of coordination I have learned that work in centers of coordination puts demands on the tools and technology used. There must at no time be any ambiguity about the current situation. However, the vast number of sources for the data; screens, printouts, reports, colleagues, etc demand much of the people working in this setting. They must not only know how to read the information they get, they must also know how to combine different sources and know when they have sufficient information for making a decision. The work in centers of coordination thus relies heavily on rules and regulations so that the actors know what to expect. Everyone needs to work in a similar way for the participants to be able to *predict* and *interpret* what others will do and are doing. The work is also characterized by a certain amount of redundancy; checking information and checking again, not only doing the work within the task description but also looking out for how the colleagues are doing.

In a study of air traffic control in Copenhagen that I participated in (see Berndtsson & Normark, 1999; Berndtsson & Normark, 2000; Normark, 2000), it was learned that the controllers used an array of tools and practices that together make for effective controlling. The air traffic controllers cannot physically control the aircraft in the sky, but give directions

through radio communication with the pilots. The controllers are directing the aircraft (giving instructions about heading, altitude and speed) continuously in order to keep safety distance between them. The airspace is divided into sectors, and thus is also the responsibility for the aircraft, since they are in and move between the sectors. In order to keep track of the current flights in a sector, each flight is represented by a paper flight strip, which is hand annotated with the latest orders given. Since the aircraft usually crossed many sectors during a flight, coordination of the current activities to the adjacent sectors that will be responsible for the aircraft in a near future is essential. One way to distribute information about the current state to a colleague in an adjacent sector is to broadcast the activities through a video/audio link. There are several important features of this system: unobtrusiveness, it communicates a manageable amount of information, supports real time view of the state and it make use of artifacts that are already there for local communication. Both the creative solutions to coordination problems as well as the need to reduce verbal communication to less time-consuming actions are typical for the centers of coordination settings. The verbal communication is "saved" to the cases that are out of the ordinary.

3. Concepts of work and technology

In this chapter some concepts and distinctions of work and technology are discussed. Besides being useful for my analysis, these concepts could also be viewed upon as a suggestion for the earlier mentioned collection of concepts that ethnography could contribute with as reference points in design projects.

Tool-mediation and double level language

Most activities that we do involve different tools that we are dependent on. It is therefore no surprise that coordination is often helped by and dependent on tools. Tool, as the word is used in this section, has a wide range of meanings, any object, artifact can be used as means for coordination. A basic assumption is that manipulation of tools are explicitly or implicitly used for coordination. Pål Sørgaard gives an example on how an object mediates coordination (Sørgaard, 1988):

"A simple example is the way two people carry a table. A part of the coordination may take place as explicit communication, for example in a discussion of how to get a table through a door. When the table is carried, however, the two people can follow each other's actions because the actions get mediated through the shared material. This coordination is not necessarily explicit."

Typical tools that we use for explicit coordination are those in which we can imprint a message; write or record them. There are a vast number of tools that surround us, which communicate written messages, for example the sign on a door saying Occupied. An implicit coordination tool can be any object that through its placement, alignment or modifications

conveys a message. These messages are strongly connected to the context. A simple example: to keep our seats in a train we might put our jacket in the seat when going to buy coffee in order for others to see that this seat is taken. We understand this cue not because it is a part of a universal language for keeping our seats, but because we are familiar with the activity of going by trains in the western societies. If someone would remove our jacket and take our seat, we are rightfully in the position of being allowed to ask that person to move.

Since the implicit cues that we use for coordinating with people around us are so dependent of the context, it can be assumed that every professional context has its special cues that others, laypersons, would not at first sight understand. When planning a redesigned system for work, I believe it to be important to know what kind of cues that the professionals depend on. Zuboff (1988) gave an example where office workers made little holes in the cubicle walls in order to have some awareness of each other's activities. The walls were blocking the informal contact that they felt was important. Robinson (1993) writes:

"Implicit communication can only happen when the participating actors are able to *maintain* an evolving set of rules, understandings, and expectations about the meanings of actions, signs, and changes of the common artifact/.../." (p. 195)

In order for one to use, say the jacket of your colleague on a chair, as an indication that s/he is at work, it has to be a recurring phenomenon. The familiarity with an object and its usual placement (or the missing of it) that is used for a cue is necessary for implicit communication to happen. Again, the inside perspective of a competent member becomes the possibility to interpret information.

Coordination mechanisms and articulation work

The quote above could very well be a starting point for the research done on Coordination Mechanisms. Schmidt is the researcher that most often is connected with these theories. He and his colleagues have worked on a conceptual framework for coordination mechanisms (Schmidt & Simone, 1996). They use the concept of CM not in the everyday sense but have an intricate description of the phenomena that surround artifact use for coordinative purposes. Schmidt and Simone make 26 propositions based on knowledge from a set of field studies on the use of artifacts for coordination purposes. The question is how to reduce the complexity in cooperative activities. In the framework, several analytical distinctions and concepts are defined, many of them related to *articulation work*.

Proposition 1 makes a distinction between cooperative work and articulation work:

"*Cooperative work* is constituted by interdependence of multiple actors who interact through changing the state of a common field of work, whereas *articulation work* is constituted by the need to restrain the distributed nature of complexly interdependent activities." (p. 158 *ibid.*)

Articulation work occurs when the *field of work* is *complex*, and there is a need to communicate activities to reduce the complexity. The field of work is the alterable material (concrete or virtual) that people at a work place is working on. That could e.g. be a database for bookings

that several persons work on. Complex work situations are those where the current state of the field of work is (or may be) ambiguous and the field of work is dynamic and unpredictable. Schmidt & Simone (1996) writes:

"To deal with this source of confusion and disorder, individual and yet interdependent activities must be coordinated, scheduled, meshed, integrated, etc. – in short: articulated (the word 'articulated' is used in the sense of 'to put together by joints'). That is, the orderly accomplishment of cooperative work requires what has been termed *articulation work*." (p. 158)

In **Proposition 2** Schmidt and Simone point out that articulation of work and cooperative work are recursive, in that the way of articulating work may be in need of cooperative negotiation, and that the result of it may be in need of articulation. Take for example a classroom with traditional settings for learning, teacher in front and several pupils facing her. In order to make sense of what is taught in the group, there is a need for attention. Hence, the rule of lifting one's hand before speaking. The cooperative work consists of various contributions to the topic of discussion, whereas the articulation work is the lifting of the hand for reducing the complexity of several people in a room that are speaking jointly. A recursion of cooperation and articulation work would in this case be initiated by a suggestion that there should be a change of the articulation of wanting to say something, i.e. lifting ones hand. One pupil suggests that the teacher's role of appointing the current speaker should be removed and there should be a stick to exchange between the talkers and move the word around with. In discussing this, the class makes a cooperative effort, which later may result in a change in the articulation of wanting to speak.

Proposition 3 deals with the role of the artifact in cooperative work (ibid.):

"In cooperative work settings characterized by complex task interdependencies, the articulation of the distributed activities requires specialized artifacts which, in the context of a set of conventions and procedures, are instrumental in *reducing the complexity of articulation work* and in alleviating the need for *ad hoc* deliberation and negotiation." (p. 162)

The coordinative functions of artifacts are a reportedly very important aspect in complex work settings. To be able to see and communicate what others are doing and what the states of different tools are, is crucial for the coordination of work, (see e.g. Berndtsson & Normark, 1999; Harper & Hughes, 1993; Heath & Luff, 1992; Suchman, 1993b, among many others). The important issue here is dealing with the complexity of work. As reported in the studies mentioned, numerous ways to reduce complexity have been developed. In the case of air traffic control, we have studied how a simple surveillance camera system is used for coordination of the important flight strips in the rack, see Berndtsson & Normark (1999). By using the cameras and monitors, the distribution of information is reduced and the controllers have the information at their own work suite. The amount of relevant information that has to be coordinated through – what is experienced as - obtrusive phone calls is reduced.

In **proposition 4**, Schmidt and Simone suggest that an *artifact* with a *coordinative protocol* is a coordination mechanism. They emphasize that the coordinative functions of the artifact should in some way include a protocol, i.e. a plan. They illustrate a coordination mechanism by the bug report forms used in a software development project:

"On the one hand, we have a *coordinative protocol* in the form of a set of agreed-to procedures and conventions which, to competent members of the ensemble, stipulates the responsibilities of the different roles, the possible classifications of bugs, the intricate flow of forms, acknowledgments, reports of bugs corrected, etc. On the other hand we have the bug report form as an *artifact*, i.e., as a distinct and persistent symbolic construct, in which the protocol is imprinted and objectified." (p. 165)

Awareness - mutual, general, peripheral, social...

Awareness means to sense others activities, either through hearing, seeing (and possibly smelling, feeling and perhaps even tasting), draw a conclusion, and make use of that in one's own activities. This could either be directly, for example when you see them passing in the corridor, or it could be indirectly through for example seeing that someone is uploading a file on a shared computer system. Awareness of each other's activities is essential for coordination of work, so it is interesting to see how tools are used for improving awareness of the general state of affairs. Consider the example that Zuboff gave about making a hole in the cubicle wall; the office workers were not able to see details of what their colleagues were doing, like seeing what they read, but they got a general idea of what was happening, possibly from the posture of the person they looked at and their knowledge of how the cubicles were equipped.

When working in the same office, for example, we can achieve awareness of each other's activities from the shared physical environment; steps in the corridor, talk heard through walls, printers printing, etc. The real challenge is to support distributed work. Hollan & Stornetta (1992) state that "there is a predictable fall-off in likelihood of collaboration between two researchers" in distributed settings, even if they have the same interests and are at the same organizational level. A likely reason for this is said to be the lack of informal interaction that creates the common ground for the maintenance of a relationship. A basic presupposition is that it seems to be more difficult to get an informal relation in a distributed setting compared to a co-located setting. How can we record the steps in the corridor, the schedule outside the meeting rooms, the informal encounters in the hallway, etc so that we can get the same kind of awareness at a distance? Schmidt (1998) writes:

"From a CSCW perspective, the crucial point about the concept of mutual awareness is to understand how mutual awareness is produced, i.e. (1) how information pertinent to mutual awareness is provided and acquired by members of the cooperative ensemble, and (2) how the characteristics of the work setting constrain and afford the provision and acquisition of information pertaining to mutual awareness."

As Schmidt & Simone (1996) point out, one aspect of coordination work is that, as a competent member, you actively provide the cues that you know your colleagues are

interested in. It is part of being a competent member to be able to predict what others will be interested to know. A quite common example at least in the academic world is keeping your office door open or closed. It is often a sign of the level of privacy that is requested by the person in the room.

Juxtaposition of sources

A professional knows how to combine information in order to interpret the current situation.

"The mutually constitutive relation of actions and their environments includes the fact that accounts of activity are themselves crafted from the juxtaposition of observable features of embodied actions with phenomena selected from the scene in progress /.../." (Suchman, 2000)

This concept is especially useful for control room work, where the professionals have to make fast decisions based upon different pieces of information. How are the pieces combined? How do they make sense of the information? Suchman reports from a study of luggage handling at an airport:

"A central finding of our analyses concerns the extent to which the work of Operations involves the assembly of knowledge about past, present, and future events through the juxtaposition and relationship of a diverse range of technologies and artifacts. Access to information and its timely communication to relevant others involves interaction not with a single technology but rather with multiple technologies (e.g. forms, computer screens, video monitors) held in relation to each other and read off in ways specifically structured by the task at hand." (p. 47 Suchman, 1993a)

From what is known about control room work, the time constraints are an important shaper of the work setting. The possibility to get information, at the right time and at the right time and to be able to integrate it with other sources is important.

Boundary Objects

What does the boundaries of an object represent? In order for an object, or anything at all, to get a specified boundary, there needs to be some kind of categorization. Take a chair for example. In some contexts it is an item for sitting. If the same object would be used for children's building of an indoor tent, it would be a tent pole. Depending on the context, the boundaries for what it is and does change.

"...we define boundary objects as those objects that both inhabit several communities of practice *and* satisfy the informational requirements of each of them. In working practice, they are objects that are able both to travel across borders and maintain some sort of constant identity. They can be tailored to meet the needs of any one community /.../. At the same time, they have common identities across settings. This is achieved by allowing the objects to be weakly structured in common use, imposing

stronger structures in the individual-site tailored use. They are thus both ambiguous and constant; they may be abstract and concrete." (Bowker & Star, 2000, p. 16)

The boundary object and the discovery of boundary objects at a workplace have serious consequences for the understanding of work and design of new artifacts. If an object (the same object) is considered a chair by one part and a tent pole by another, many confusions may arrive when redesigning the object.

"Such objects have different meanings in different social worlds but their structure is common enough to more than one world to make them recognizable, a means of translation. The creation and management of boundary objects is a key process in developing and maintaining coherence across intersecting communities." (Ibid. p. 297)

Of course, maintaining an object within an organization that is categorized differently by different members requires a lot of effort. Values, names, use, meanings and priority are all things that come with an object getting classified. Managing these different views creates a further complexity in between colleagues, groups, departments or organizations.

4. Summary of the theoretical discussion

Computer Supported Cooperative Work is a research area that among other things has put focus on work, coordination and technology use. Often, these qualitative studies have been referred to as workplace studies. My interest in the use of technology for coordination of work is inspired by this tradition. Coordination is defined as joint work, where the actors are immediately dependent of each other. This research does thus not focus on, or analyze, any organizational work arrangements. A special interest in the CSCW area has been control rooms, where the shift between the normal state and the emergent state puts special demands on the technology; nothing must go wrong in these shifts. I have found the term Centers of Coordination (Suchman, 1993a; 1993b) useful for understanding the character of the work going on there. The need for being able to check and have overview, not only of ones own work, but of the colleagues as well, puts the technology to the test. How do the actors maintain an awareness of the work going on and when someone needs help?

I have in the chapter about workplace studies tried to outline the theoretical, mostly sociological, background to these kinds of studies. It has been interesting and necessary for me to get a clearer understanding of workplace studies through studies of ethnomethodology. The general question of how people achieve an interactional effect is in my opinion highly relevant for the studies I have done at SOS Alarm. In order to indicate interaction without having to spell it out, what Robinson (1991) calls implicit language, one has to manipulate the object so that others can see that something has been done. When it comes to the design of what I call practical objects, like a hotel key rack where the keys are in or out, the possibilities and constraints of the material are rather intuitive. When it comes to

abstract, intellectual computerized material like in the CoordCom system at SOS Alarm that will be discussed later on, it is rather less intuitive.

The method of studying workplaces is often ethnography. It generally means that the researcher does observational studies in the setting of interest and writes about the practices and skills of the participants. Ethnographers are sometimes "accused of" a fascination with the current practices that does not lead to formal demands for a new design. This is the wrong interpretation of ethnography. However, better ways for involving ethnography in design projects are needed. Different suggestions have been made on how ethnographies can be more effectively used in design projects. One has been that we could start to assemble reoccurring issues, as a framework, which I think is an interesting approach (Bannon 2000). I have tried to select the concepts that I have found interesting in the workplace studies I have read. These were tool-mediation and the double level language, coordination mechanisms and technologies for articulation work, possibilities for awareness of the activities of others, how sources and tools are juxtaposed in order to take a decision, and how objects' boundaries are differently understood by different actors. Other reoccurring concepts in workplace studies are part of the ethnomethodological "framework" like "accountability", the methods that actors' use to exhibit the rationality of their activities.

5. The SOS Alarm study

An important reason for us getting into SOS Alarm to study coordination was that they aim to develop a totally new system, which will be finished (according to the plans), in 2005. A main change in this new system will be that the 20 centers will share the case database so that any center can receive any call and redirect the case to the right center. At least in theory, this means that the call can be handled at the center that is least busy, regardless of geographical proximity to the caller. This opens up a whole set of questions, e.g. what will happen to the overview and the knowledge that is situated within the room, what will happen to the local knowledge, what will happen to the coordination when the coordinating operators have less awareness of each others activities, etc. I have had this change in mind when analyzing coordinative actions inside and outside of the computer system.

In this thesis I from time to time describe the operators' work as complex. Since the ordinary, everyday meaning of complexity is somewhat vague (nearly all work is "complex") I would like to make a definition of what I mean. I use it for what Schmidt (1993) calls apperceptive (interpretive) complexity. This is when the overview can never be complete and that there might be unknown states. Possible states are infinite so that work cannot be (totally) predicted. Schmidt quotes Woods (1988) from his book *Coping with complexity*:

"When there is high uncertainty, available data can be ambiguous, incomplete, erroneous, low signal to noise ration, or imprecise with respect to the state of the world; the inferential value of data can vary with context; future states and events are not completely predictable."

The complexity of the operators' work is also what Schmidt (1993) calls structural; when the work includes different interconnected tasks so that a failure or problem might have several consequences (not always foreseeable).

The complexity of the work thus is the unpredictable nature of it and its interwoven tasks.

SOS Alarm AB - general introduction

The 20 SOS Alarm Centers in Sweden have certain security rules and regulations. Besides from having an appointment, you need to be let in at the gates and then also let into the actual control room. There is a shift leader that is responsible for the operative work that handles guests. On agreement you are allowed to sit beside an operator and listen in to the incoming emergency calls, but you have to sign a professional secrecy agreement first. The control room is shelter classified and has large concrete blocks that can cover the windows. The centers differ in size. At the three larger centers there are 8 operators or more working at daytime while in the smaller centers there are 2-3 operators working together. The operators sit at positions equipped like this (it differs marginally between the centers):

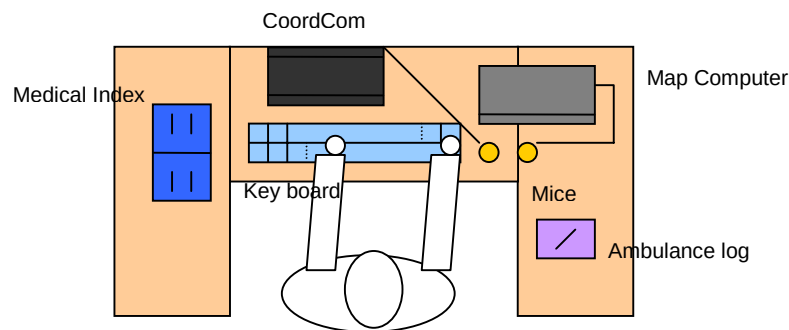


Figure 1: Malmö position

The operators' main tool is the computer-aided dispatch system called CoordCom. Through the system they handle all communication: telephone, fax, mobitex, radio. They also do all documentation in it. The system is connected to different databases, both the local ones: all action plans, ongoing cases, resources, contact information and also to a central database: the telephone subscription record that generates addresses from the caller. The CoordCom is handled through a specially made keyboard that has selected functions to suit the system, for example to answer a 112 call. There is also a mouse and the operator's use a headset connected to the computer to communicate.

At some centers in Sweden there is a Map Computer in use as well. This computer is mainly used to track the units in the area through the GPS system in the cars. The system can also be used to look up addresses, but is usually considered to slow for that.

An incoming 112 call is answered with "SOS 112, what has occurred". That is the first thing the operator tries to decide. Is it a case for SOS or not? What kind of priority should be assigned to the case (1-4)? Do I need to ask a fellow operator to listen in to the call? As we will see from the Stockholm and Malmö cases, this division of labor is handled differently. While deciding what kind of accident it is, the operators work with CoordCom to document and label the accident. The second kind of information is the address of the incident. After

putting in the address, a colleague can start dispatching (if the case is priority 1) while the receiving operator can collect more information about driving directions, development of the accident, etc from the caller. When the call is finished, it is the dispatching operators' responsibility to handle the case. S/he chooses among the resources that are suggested by CoordCom, based on proximity of the ambulance station. After calling the ambulance verbally, the operator sends out a mobitex message, a text message that gets printed out in the ambulance, containing the case information that was entered into CoordCom. The mobitex system is also used to send automatic status reports from the paramedics to the SOS operator. While the ambulance is turning out, there are several kinds of status reports sent. These are for example status U - we have received the call and are on our way, S - soon arriving, F - at the accident site, L - the patient is now in the ambulance. (The letters are the beginning of the Swedish words for what they are doing, e.g. U = utryckt.) The dispatching operator then follows the advancement of the ambulance or rescue vehicle through these status reports. S/he may also help coordinate information between different vehicles, but the dispatching and monitoring can be handled without any verbal communication between the operator and the units.

The sites

In Malmö, nine operators work at daytime. Four of them work with ambulance dispatch, two of them work with rescue service, two work with commercial services and one works as shift leader. All of the operators can answer 112 calls. In Malmö the GPS system in the ambulances is fully developed, i.e. the operators can follow the movements of the ambulances on the map computer. The table that the operators sit at can be risen up so that the operator can work standing up. This is useful for the operators but can block the sight somewhat.

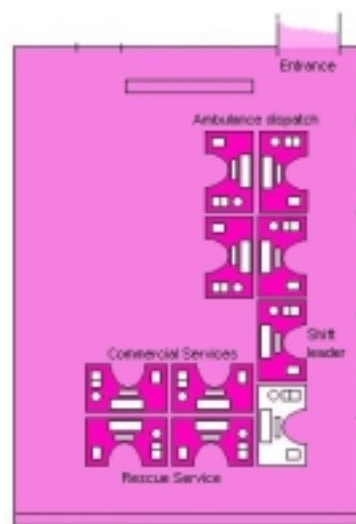


Figure 2: The position of the tables at Malmö SOS center

Between the ambulance dispatch and the rescue service operators the distance is about eight meters. That means that they cannot talk straight out, they have to communicate through the CoordCom system. All the operators do have the possibility to get eye contact, which they usually try to get when they communicate through the system.

The four ambulance dispatchers are divided into three geographical responsibilities and one coordinator, who also works with Malmö cases. Malmö has a large regional hospital, which means that there are many transports to and from it. In order to work as a dispatcher, according to the operators, one needs experience and a good knowledge of the geographical area. The coordinator position demands a specific knowledge of Malmö city and a vast experience of dispatching. Typically, operators who have worked as ambulance drivers in Malmö work in the position. (Both ambulance driving and nursing are common backgrounds for the operators.)

In Stockholm, about 15 operators work together at daytime. About six or seven work only with receiving 112-calls, three operators work with ambulance dispatch, one operator is shift leader, three operators work with commercial services. In Stockholm you need to be trained as a staff nurse (or practical nurse, undersköterska) in order to answer 112 calls. There is also a physician working at the center during daytime.

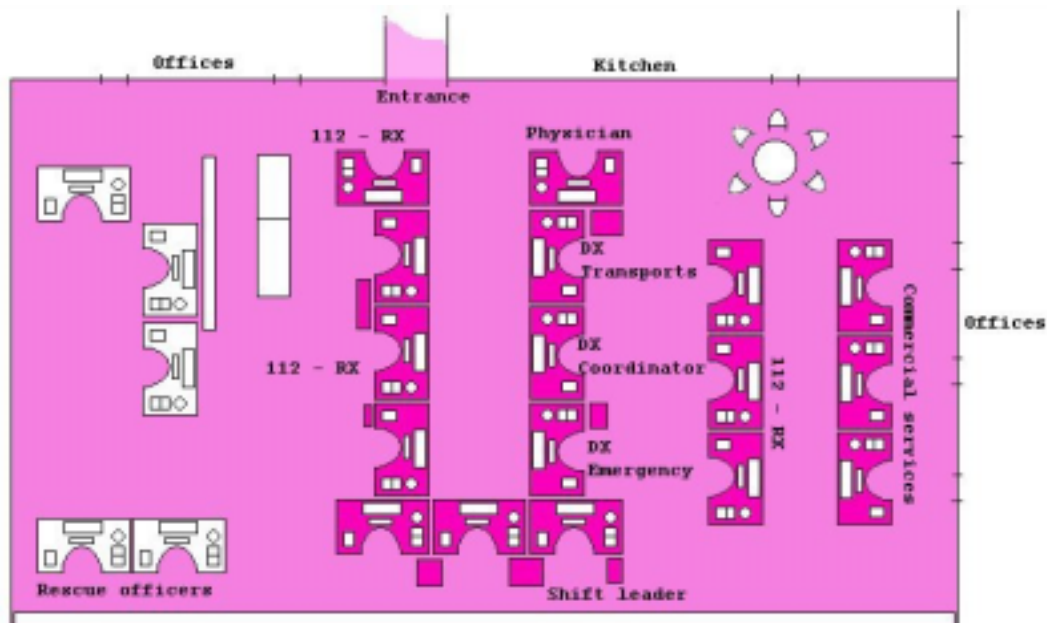


Figure 3: Stockholm's SOS center where about 15 operators work altogether work at daytime (these positions are in a darker color)

The Stockholm center is quite large, about 25x15 meters. It is placed 37 meters below ground. In the left corner the rescue officers on duty are working, they belong to the rescue service staff that is located in the same building. The Medical Information Service (Sjukvårdsupplysningen) earlier manned the other free tables to the left. In Stockholm, the

operators handle no rescue service cases, these are, as with cases for the police, directly forwarded to the rescue service's own call centers.

At the Stockholm center, there is a static division of labor between the 112 receivers and the ambulance dispatchers. Since it is rather noisy in the room, the 112 receivers and the dispatchers do not so often overhear each other's work. However, they do see each other and thus can judge if the other is busy or not.

In chapter 8 I present two typical authentic cases from my study in Stockholm. They give a hint of the problems and varied nature of the work that the SOS operators are doing.

Project description and methodology

The SOS Alarm research project was initiated by the Work Practice Laboratory at Blekinge Institute of Technology in Ronneby and SOS Alarm. The project aimed at finding out how the operators use the technology to perform their tasks and how it could be supported in the future.

My field studies began in June 2000 at the Malmö SOS center. Together with Jenny Lundberg and Mårten Pettersson from Ronneby we continued the study until November. I mainly made observations in Malmö and Stockholm, while Jenny and Mårten mainly made observations at the Växjö SOS center. Since I live in Stockholm and had to travel to Malmö, I usually went there for concentrated periods of 1-2 whole days, about 15 times.

During the project we went on several study visits besides being at the different centers. I took part in the visits to Jönköping, Växjö and Borås. Borås has a special operator position that is really a part of the Gothenburg center. This one operator that works in Borås is there to cooperate with the fire rescue department for the east area of the Gothenburg responsibility area. Another study visit was to the Rosersberg Rescue Center located in a suburb of Stockholm, where we took part during a day when new SOS Alarm operators were educated. Further, we went to study the computer systems at other kinds of centers, namely the Øresunds bridge and the Police headquarters in Malmö.

We held two workshops, one in Malmö where we discussed problems of today and ideas for the future together with the operators and one in Växjö where we discussed possible new interfaces and functionality for a new way of performing the work in-between the centers. We were also a part of the reference group that SOS Alarm head quarters had put together for the first stages of the process of developing a new support system. The project resulted in a report with four parts, Helgeson et al. (2000) that were delivered to the SOS Alarm. During the spring of 2001 I continued my field studies at the Stockholm SOS Center until the summer, but with less frequent observations than earlier (about 7 visits).

The method I used in my studies was ethnography. Basically, ethnography is used as a method to understand how an activity is performed in a certain setting, and the outcome is a written account. I have already discussed some of the fundamental assumptions behind ethnography in the theoretical part so I will not explain the method here in detail again but describe what I did in my field studies.

The main data collection method that I used was extensive notes, but I have also taken photographs and collected printed information. At a few occasions we made video recordings. In one longer case, the childbirth case that is presented in this thesis, I got access to the transcripts of the recorded conversation between the operators, the paramedics and others. A shift leader extracted the tape from the SOS own recordings and it was time-consuming work since the calls are not tagged on the tape.

The SOS Alarm setting is difficult to study for two reasons; one is that secrecy agreements put special demands on the recording possibilities and another is that there are (for my research interest) very many sources to take account of; talk in the room, conversation with different rescue services through telephone, radio etc. and also work and coordination through the networked CoordCom system. An utopian data collection method would have been to collect a synchronized version of these three parts.

Recordings also to some extent presented an ethical problem for me; how would the emergency callers feel if they knew that their emergencies were recorded and used for research at a university? Although this has been done before, I personally felt it to be somewhat problematic. Because of my focus not being specifically on the conversation, and the problems that surrounded other data recording techniques, I decided to mainly use notes as data for analysis.

When doing ethnographic studies one never knows where to stop. The most revealing case may be around the corner if you remain a little longer. A moment's lack of attention, information that you just missed, may have included just that interesting piece that would have rendered a different analysis. However, I believe that the account of SOS operators' work and interaction with technology in this thesis is rather accurate. To make sure, I have let some people that were involved in the project and that are working at SOS Alarm read and comment on what I have written.

6. Domain related research

In this part, I briefly describe the work and conclusions of other researchers analyzing emergency dispatch work. I will begin with researchers that have done a conversation analysis of the initiation of the emergency case, namely the first phone call from a citizen. Further, researchers that have analyzed the construction of a case, mainly after the phone call and the interaction between the operators during this phase. Thirdly, another group of researchers has studied the managing of resources between operators when the case is established.

In Whalen, Zimmerman, & Whalen (1988) there is a tragic authentic case analyzed. The miscommunication between the caller and the receiving operator and emergency nurse leads to the death of a woman. Whalen et al. make an ethnomethodological analysis of the conversation, in order to understand how the interaction failed. This is an exceptional case, which made top headlines in the city where it happened. The study points to the important fact that it is the conversation in the emergency phone call that is the foundation for the implementations and decisions that the operator makes and thus what resource that is sent out.

The first problem in the case arises in the following section (only conversation included):

D=Department Desk Operator, C=Caller

D: Fire department

C: Yes, I'd like to have an ambulance at forty one thirty nine Haverford please

D: What's the problem sir?

C: I don't know, if I knew I wouldn't be calling you all

This part very well describes the nature of the call. The caller thinks that he is calling to order an ambulance and he gets angry when he has to explain why. The attitude and cursing which occurs later on in the call makes the nurse angry and she does not take him seriously. The conversation finishes when the caller gets tired and does not want to answer questions anymore. When one reads the transcript it turns out that the caller has given information that shows the life threatening situation but presented it in such a way that the nurse does not react upon it. Summing up the important cues the nurse got (I have removed the pauses and the transcript just shows what was said):

Line	Text
11	It is my mother who needs the ambulance
50-51	she is having difficult in breathing
61-62	you can't speak with her. She's seems like she is incoherent
77-78	this is a life threatening situation
89	she could die if you don't hurry
95-96	she is in there and can't breathe
99-100	she is having difficult in breathing, she cannot talk
129	she is incoherent
136-137	she is incoherent, she cannot talk
139	she cannot talk at all
150	she can't talk
151	she can't talk

(For a further description of this case, se the article, Ibid.)

This incident generated a lot of media attention. The question that everyone asked was how could this happen? Whalen et al. use this question as a starting point for their analysis. It seems that the expectations on information to be given are so different that the nurse and the caller cannot understand each other. The nurse does not seem to get a "label" of the accident at the times she asks for them. A label is however provided at other places in the talk, e.g. line 50-51 above. Whalen et al. write:

"Our investigation revealed that the participants had rather different understandings of what was happening and different expectations of what was supposed to happen in this conversation. Over the course of the interaction the talk of both caller and nurse-dispatcher (and her supervisor) operated to extend and deepen this misalignment. This misalignment contributed in a fundamental way to a dispute that contaminated and transformed the participants' activity: the eliciting and giving of information concerning the condition of the caller's stepmother was displaced with the activity of arguing." (p. 358).

The emergency phone call is not a neutral exchange of facts. The account of the caller is affecting the judgment of the receiver. The emergency phone call made by a layman is perhaps the weak link in the emergency dispatch work. However, the receiving operators must not get "blind" for the facts that are conveyed. It would be interesting to know how much the documentation tools that the nurse used added to the special order she needed

things to be in. The case resulted in the nurse being fired because of her failure to receive information in the wrong order.

This research aimed for discussing the incoming emergency call while the next study concerned the receiving and establishing of a case in the control room.

In his Ph.D. thesis, Henrik Artman (1999) focuses on cognition in dynamic command and control centers. In understanding the coordination in these centers, Artman draws on the Distributed Cognition perspective where representation, task and activity are analyzed separately. He takes great interest in the parallel and yet group relevant work and decision making that take place. Communicating parallel processes, what each participant does, requires extra focus informing on all things that are going on. Access to, and presentation of, information is thus crucial for supporting awareness. One point that Artman makes is the importance of the common room, and the possibility to support situational awareness by visual cues.

One of the potential problems with the common room is, according to Artman that too many people tend to get involved when there is a problem "...everyone throws themselves over the nearest problem at hand and try to solve it together. This often results in many people getting involved in a rather simple problem and add more perspectives than is really needed, which makes the problem-solving slower than if one person would have taken care of the problem" (p 118, Ibid.)

Artman also warns about the effects of new technology, although it is needed:

"New technology is necessary and often brings positive effects /.../. But new technology may have short-term positive effects but in the long run destroy social practices that have been built up by generations of operators." (p 119, Ibid.)

Although this is not a surprising opinion, I agree that the social practices have to be considered and to some extent evaluated so that new technology does not hinder informal trajectories of information. People will always use the possibilities and constraints of the situation as resources to accomplish interaction, and they are flexible. However, it is important to understand what these possibilities and constraints of the setting offer and to what extent they are used and relied on.

Artman & Wærn (1999) take an interesting focus on the construction of cases at SOS Alarm. They write that "this /data/ shows how the knowledge and interpretation processes are distributed between the operators." In that sense, Artman and Wærn see emergency case establishment as a shared task. This means that deciding how to categorize an accident and what resources to send is made through a negotiation amongst the operators, even if it at first sight look like one operator's responsibility. The conclusion suggests that operator work, however individually divided, is largely dependent on others in order for it to run smoothly.

The last study is focused on the dispatch process and the interaction between the dispatcher and the supervisor and the technology.

Martin, Bowers, & Wastell (1997) have done a study of coordination between the dispatcher that is responsible for send out the ambulances and the supervisors that oversee the work of the dispatchers at an Ambulance Control Center. They were interested in the

way the actors organize the distribution of ambulances; the methods that they use in order to achieve interaction in this context. One aim for their study was to compare this site, where computers are a part of work practice, with the control room studies made in London Underground, (Heath & Luff, 1992), and Air Traffic Control, (Hughes et al., 1988). The results in these previous studies pointed towards importance of the affordances of easily manageable non-computerized materials used for coordination. Martin et al. wanted to show an example of similar work successfully, as they had noticed, executed with the help of computers.

A difference from the study presented here is that in the study of Martin et al., there is almost no communication going on between the receiving operators and the dispatchers. Another difference is that there is no role such as the supervisor in the Swedish setting. The Martin et al. article is not very clear on the general division of labor between these two roles, saying mainly that the supervisor manages the dispatch and keeps an overview of the whole area. Even though one of their conclusions is that the division of labor is an ongoing process (so-called working division of labor), it would have been interesting to hear more about their formal task differences.

An interesting part of the paper concerns the contingencies that the dispatchers face when choosing a vehicle. When looking superficially at the work that they do, it seems that grabbing the nearest available ambulance would be a suitable practice. However, there are a number of things to consider: work hours, meal breaks, vehicle equipment, maintaining an even distribution of vehicles ready to go in the areas, etc, etc. Martin et al. make the point that since these contingencies may be significant, there is a greater need of information than just knowing *where* the ambulance is geographically located. With support from their study, Martin et al. make the conclusion that a GPS map, following the geographical locations of the cars is not very useful and time consuming. This is because it is only possible to extract part of the information that is needed when deciding which vehicle to choose and the map itself does not give the piece of information fast. The operators have to search for it e.g. "where do I have car x...oh there it is...and how far is that to Z road...perhaps car Y is closer". Thus their point is that it takes time to look the information up on the map and it does not include all contingencies that will be used for the final decision on which vehicle to choose.

In the end of the paper, Martin et al. (1997) give the outlines for a further development of their study in making a prototype of a new system. They are guided by four conclusions:

1. Contingencies are interactionally managed (negotiation of choices). This is a similar conclusion to the one made by Artman and Wærn; interaction, rather than individual work accomplishes the choices and categorization work.
2. Division of labor is continuously worked on.
3. Representations are work-oriented. (It is not always the most advanced graphical representations that are useful. In time critical settings, simplicity and the possibility to perform a task fast are the two main foundations for a good tool.)

4. HCI is a public phenomenon (design for third parties). When the dispatcher is carrying out a task, the supervisor can understand his/her turning the head towards the information that needs attention. By seeing what the operator is attending to, the supervisor can help without the operator asking for it. The question Martin et al. raise is thus how well a tool helps others than the immediate user to understand what is going on.

7. Work description and coordinative concerns

Beginning with an overview of the main tools that the operators need, this chapter continues with a description of how a case is handled. It ends with a discussion about coordinative concerns on the part of the receiving operator and the dispatcher.

The computer system - CoordCom

CoordCom is the computer system in and through which the operators communicate, monitor and document in the cases. It is the computer system towards which the operators mainly orient. CoordCom is handled with a specially made keyboard that gives direct access to certain functions (e.g. answering an 112 call). The database in which the cases are stored is local, i.e. each center has its own database and cannot access cases documented at other centers. There is no connection between the networks (the centers can thus not communicate data between themselves). The CoordCom supports all the different ways of communicating; through telephone, radio, mobitex (a text-based communication system used between the units and the SOS central and for automatic alarms), and fax. It also supports making records of the different kind of cases, both emergency and commercial alarm service, that SOS is handling. This documentation of the cases is stored in the local central today, but the idea is that all the centrals should be able to share the information and thus be able to work on each other's cases in the future. The design of CoordCom is focused on the incoming calls. For dispatching and monitoring the ambulances and rescue units there is a "map computer" that shows the units with help of a GPS system that indicates where the

ambulance units are on a digital map. This system is very useful both for giving driving instructions and to pick the ambulance closest to an accident.

The operators log in depending on the tasks they work with. By logging in the operators filter the information that they get. An ambulance dispatcher, for example, does not see the incoming 112 calls. The login is personal as well, so that each action is stamped with the signature of the operator that performed an action in the system (e.g. made a call or received status reports from the units).



Figure 5: The CoordCom with its special keyboard

The CoordCom interface is divided into two parts; the telematics part and the case part. In the telematics part all communication (radio, telephone, and text) is handled. In the case part, all information concerning the cases is handled. More about the CoordCom interface follows further on in this chapter.

Maps and the Map computer

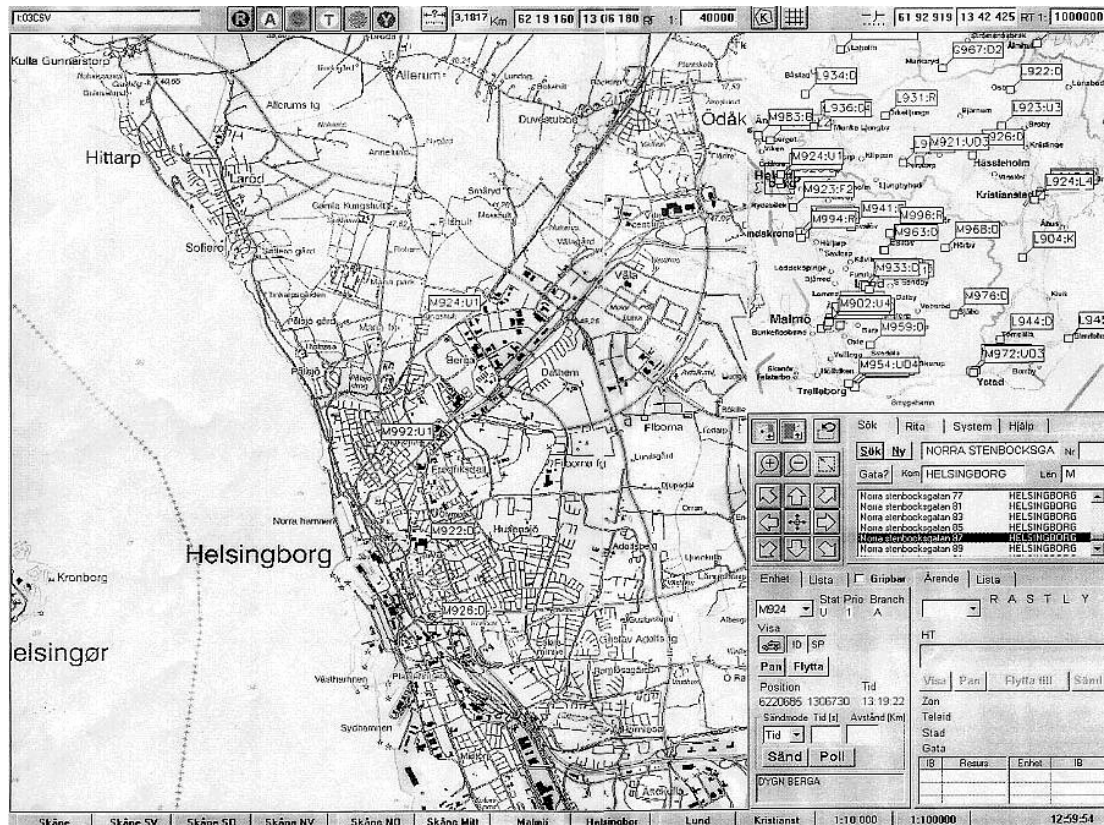


Figure 6: The map computer where places can be found and ambulances monitored.

The map computer is mainly used in Malmö. In Stockholm they do have the system, although the units do not have GPS so the system cannot be used to trace them. The system can also be used to find addresses, although it is slow. The interface consists of three fields, one that shows the current area of interest in chosen level of zoom (to the left) and one, which shows the whole area (in this case Skåne, to the upper right). There is also an area to the right below where places can be sought and other information managed. The operators can for example find out the geographical coordinates for a certain map square. The operators follow the units (the squares on the map above that contain numbers like M924) on the map computer. By following a unit, they are able to inform other teams where the unit currently is and approximately when it will arrive. By using the map they can also find out the exact coordinates of the accident, which can help the units to find it.

Information in paper sources

In both Stockholm and Malmö they have large pull-down maps that look like shades. On these maps are the important ambulance areas that the operators need to put in CoordCom in order to single out available ambulances. The maps are used to locate the incident area, to

give driving directions to the units and also to decide the area code that should be assigned to the case. The area codes are very important so that the right ambulance can be chosen. For each area, there are decisions made on which ambulance station should take the case.

Work description - incoming call

The ethnographic method with its interest in (work) practice in a certain culture, opens up for a range of different perspectives that one can take upon the work. The most obvious way to present the operators' work is to follow the development of an emergency case. However, cases are not handled sequentially but in parallel. This means that an operator does a part of a case and then hands it over and begins with another case, even if the first one is not finished. This part thus does not describe the work process but the case development.

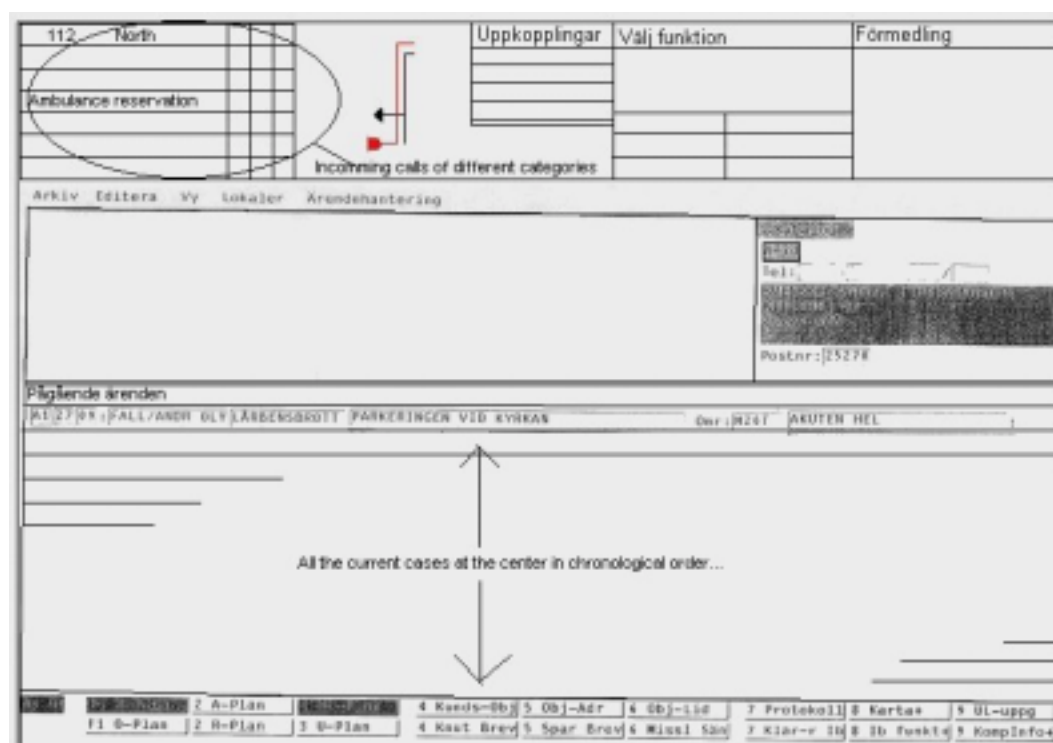


Figure 7: The incoming case view²

This is the CoordCom interface with the relevant windows that the operator views during the beginning of a call. The upper part of the interface is the telematic part. In the left corner, you can see a list of seven rows. These rows display different kinds of incoming calls or messages. This is how the calls are ordered (where no 1 is the topmost row):

1. Incoming 112
2. Co-listening requests

² The interface pictures in this chapter are based on screen dumps, which have been partly reconstructed and annotated by me.

3. Automatic alarms (e.g. from elevators or fire alarms)
4. Calls to other numbers than 112, e.g. ambulance transportation ordering (no emergency)
5. Radio calls from e.g. ambulance drivers/paramedics
6. Mobitex calls
7. Transportation of calls, internal requests, time alarms (if the ambulances do not respond in time)

The little picture to the right represents the radio and telephone lines. They show e.g. if the phone call is put on secrecy or not. This does not seem to mean much to the operators, and I think that this is a remainder from the time when 112 (or the old number 90 000) was a service managed by the public phone company Televerket. Then the service was considered more as technically connecting to the right service (e.g. ambulances) than as a part of the emergency service. In the square next to the symbol the current connections are presented.

All communication made through the CoordCom is recorded and all information that is documented is saved in case it will be needed for later analysis of the case.

The call is answered by pushing a key on the keyboard. Then the oldest incoming 112 call will be chosen. The SOS operator usually answers "Ess Oh Ess one one two what has happened" (in Swedish: SOS 112 vad har inträffat). This practice has been discussed, as the phrase is sometimes confusing for people that do not speak Swedish very well. It is supposed to make people talk directly about the accident instead of beginning with polite phrases and personal introductions. While answering a call, the operator will have a basic view in CoordCom which includes a communication part on the current incoming calls, a so-called "Basic Form" (in Swedish: grundblankett) where the phone number of the caller is automatically shown, and a list of current cases at the central. At this point, a decision-making process starts.

The first decision to be made is, obviously, if this is a case for the SOS or not. Since the change of the Swedish emergency phone number 90 000 to the European number 112, there has been a high rate of invalid calls; prank calls, wrong number, calls aimed for the phone number information service 118 118, and so on. Since 112 is an emergency number, phone calls about for example a discovered break-in in a car (not ongoing) should be made directly to the police and not to SOS. Currently, SOS estimates that almost 80% of the incoming 112 calls are without a valid reason. Another reason for a high rate of calls is the mobile phones. An accident that earlier may have generated 3 phone calls may now generate 30 calls. This has been mentioned as a problem for the operators, both for the obvious reason that it takes a lot of time, but also since the operators say that they have "lowered their adrenaline rush" for the 112-calls and are more likely to finish a silent phone line quickly, since there are so many of them. (A silent caller might indicate serious problems and should be examined further.)

The operators may further check if the case is already recorded in the system in the current-cases list (the lower part of Figure 7). While I have been studying the operators, I have never seen a redundant dispatch, although I have seen a case become initiated five times in the current cases list. Due to the list and to overhearing, the operators seem to be

able to coordinate current cases to avoid cases getting recorded more than once, it thus works as a coordination mechanism that both structures and informs the coordination.

Initiating a case

The screenshot shows the 'CoordCom' system interface. At the top left is a map with a red dot indicating a location. To the right of the map is a table titled 'Uppkopplingar' (Connections) with columns 'Valj funktion' (Select function) and 'Förmedling' (Mediation). Below the map is a section for 'Inkommande anrop av olika kategorier' (Incoming calls of different categories). The main part of the interface is a form for case documentation. It includes fields for 'Event code' (A1 27 09), 'Event in free text' (FALL/ANDR OLY/LÄRGENSBROTT), 'Driving directions in free text' (FALL Å 2 M FR STEG), 'Area code' (R:1140, R:1142, R:1143), and 'Telephone number' (25278). Annotations with lines point to these fields, explaining their function: 'Event code - chosen by the operator', 'Event in free text', 'Driving directions in free text', 'Area code - generated by the postal code', and 'Telephone number - generates address'. At the bottom of the interface is a bar with various menu options: 'Arkiv', 'Editera', 'Vy', 'Lokaler', 'Kredenshastering', '1 A-Plan', '2 B-Plan', '3 U-Plan', '4 Kust Brev', '5 Spar Brev', '6 Missl San', '7 Klar-r IN', '8 Ib funkt', '9 Inspinfo'.

Figure 8: The case documentation

During the interview, the operator documents as much as possible about the incident. The first thing s/he decides is what has happened, then where. In doing so the operators choose an "Resource Form" (Swedish: *InsatsBlankett*) to work in. The Resource Form is either of 6 types: R=Rescue, A=Ambulance, S=Service, T=Traffic accident, L=Doctor, Y=Other, and becomes attached to the Basic Form. There could be several Resource Forms attached to the Basic Form. By choosing an Event code (Swedish: *HändelseTyp*), the operator decides priority and defines the type of accident. In order to generate an "Action Plan" (Swedish: *HT-plan*) from the CoordCom system in the case, the operator needs to put in what kind of accident it is and where it has happened. In Figure 8, the event code is A1 27 09. A1 means ambulance case, priority 1 (highest priority), 27 means "Falling/slipping accident" and 09 means "broken leg". This is specified in text as well. The operator further specifies what has happened by writing: "Fall, about 2 meters from a ladder". S/he also puts in a further definition of the place where the accident has happened; "The parking lot by the church".

The area code is important, because based on that CoordCom will show a list of available ambulances in that area zone. It will not be shown otherwise. Usually, this code is generated by the address, which in turn is generated by the phone that the call is made from. The operator always has to ask about the address to make sure that it is the right one. If the call is made from a mobile phone, the operator gets no address and has to rely on the information given by the caller alone. In hesitant cases, the operators work with large paper maps where the zone borders are detailed. The map computer does not support this work.

How the dispatches are arranged is not decided by the operators, but by their "customers"; e.g. the different county councils. There are ambulance- and rescue officers in each county making general decisions for the dispatch order for each geographical area and type of accident. (If a fire occurs in C village, 2 fire trucks will be dispatched from A town and one ambulance from B city. The fire officer on duty in D city is to be notified). All these conditions are entered in to CoordCom and thus generate the Action Plan for each accident, which contains all the things that are to be done for the accident/location specified. While working on the what and where questions, the operators also write information about e.g. the person in question, earlier problems, driving directions and so on. All this information is entered in the header of the Resource Form.

Dispatch of the units

The screenshot shows a software interface for dispatching units. At the top, there are tabs for 'Uppkopplingar', 'Vali funktion', and 'Förmedling'. Below these, there's a section for 'Arkiv Editera Vy Lokaler Årendetastering'. The main area contains a form with fields for 'Id-A', 'Adr', 'HE', 'Till', 'Mann', 'Kib', 'Kom', 'Pos', 'Rt', 'To', 'San', 'Bom', 'Omr', 'Rt', 'A', 'Estet', 'Est', 'Tid', 'P', 'Omr', 'Till', 'P', 'V', 'T', 'U', 'T', 'L', 'A', 'stecning'. A list of units is displayed below the form, with columns for unit ID, status, time, priority, area, and description. A red arrow points to the unit 'M201' in the list. The bottom status bar shows various system information.

Unit ID	Status	Time	Priority	Area	Description
M201	S	1214	1	HEL	SJUKTNG
M202	S	1214	1	HEL	1 DYEN H-STN
M203	S	1217	1	HEL	1 AKUTBIL
M204	F	1212	2	M201	8 DAG
M205	U	1214	2	M201	1 DYEN BERGA
M206	D	1242	1	M201	DAGBIL
M207	R	1243	1	M201	DYEN

Figure 9: The area code generates the list of units in their area (and their current status)

In this view the operator is looking at the list of available ambulances. Each row contains information about the name of the unit, its latest reported status, when that status was sent in, priority status of that specific unit, which area it is in and a note about the type and a description of the unit, e.g. "Day" or "Transport".

The policy of SOS is to start to dispatch the units at most 45 seconds after the emergency call has been answered in case of a priority 1 (life threatening) case. In order to be able to collect as much information as possible and/or give advice to the caller to help the injured in some way, the operators collaborate. It is likely in emergency cases that the operator that answers the call is working on another responsibility area. This means that the first thing the operator will do is to make a co-listening (or listening-in) request to the operator that is responsible for the dispatches in the area. In the case of a fire, there would be both ambulance dispatches for that certain area and the rescue service. The co-listening request is done through the CoordCom, and as soon as the dispatching operator has acknowledged the request, she or he will show this by saying, "I am with you". In order for the receiving operator to inform the dispatching operator what the call is about, the operator usually repeats the information e.g. "so your father has fainted and he is not reacting when you try to shake him?". The dispatching operator can start to send out the units before the receiving operator has finished the phone call.

When the Action Plan is generated in CoordCom, it shows a list in priority order on what should be done. For e.g. a larger traffic accident or a fire in a larger industry, there may be ten or more units/persons that should be notified. This takes quite a lot of time, and the operators help each other. The operators usually overhear what is going on and if it is an emergency, the operators form temporal teams for dispatching and informing all units in the Action Plan. These teams are either initiated by the receiving operator asking "can you contact the police" or the operators around him/her volunteering; "I take the police". All the operators in the central have access to the current cases. In order to keep track of what has been done, there will be a time stamp in the Action Plan when a task has been done. E.g.

12:43:35 Send MTX to unit Z923. %LN%Z923%MTX%

The Action plans contain so called Logic Numbers for each task which are clickable and render a call or a sending of a mobitex message, etc. The first contact is made by radio. The operators inform the units briefly about priority, kind of incident and where the accident has happened. After assigning a task to a unit, the unit has about two minutes to (sign in or) acknowledge the mission. If they do not acknowledge the mission the operators will try to contact them through mobile phone or mini call. The acknowledgement by the paramedics or rescue team, can either be done by contacting the SOS by radio, calling through the telephone or by sending a text message through mobitex. From the perspective of the operators, it looks like an incoming call containing a text message with the status of the unit. About the same time as the acknowledgement is received, the operator sends a mobitex containing the header of the Resource Form.³

³ Possibly due to the relatively new mobitex system (about one year old) the operators often call the unit before sending the mobitex message or ask about it when the unit calls in. It seems to be important to be a bit redundant in the communication to make sure that information is spread properly to the units and to other operators. Another possible reason for (redundant) verbal communication may be to maintain goodwill in the relationship to especially the paramedics, which the SOS has full delegation over.

Monitoring the development

Nr	Tid	Å	Ib	Åtg.	Ex	Sgn
024	1214	A	001	Ehbet M922 har rapporterat status 5	1	00
023	1203	A	001	Ehbet M922 har rapporterat status 1		
022	1203	A	001	Ehbet M922 har rapporterat status 1		
021	1151	A	001	Ehbet M922 har rapporterat status 1		
020	1151	A	001	Ehbet M922 har rapporterat status 1		
019	1147	A	001	Ehbet M922 har rapporterat status 0	1	CJ
018	1147	A	001	Ehbet M922 har rapporterat status 0	4	00
017	1147	A	001	Ehbet M922 har rapporterat status 0	1	CJ
016	1147	A	001	Ehbet M922 har rapporterat status 0	1	CJ
015	1146	A	001	Extern info skickad till M922MTX	1	CJ
014	1146	A	001	ARENDE-MTX M922 SLÄNS922MTX	1	CJ
013	1145	A	001	Tidsövervakning har besvarats		
012	1145	A	001	Tidsövervakning har slagit för Ib:n	14	LI
011	1143	A	001	Extern info skickad till M922MTX	1	00

Figure 10: The events recorded by CoordCom in the measure list (below)

In Figure 10 the operator has opened the measurement list (Swedish: Åtgärdslista) that shows all activities happening in the case according to the CoordCom system. This is e.g. when text is updated or added in CoordCom, e.g. if a driving direction is added. In the list above there is mostly the communication and status reports by the ambulances that are recorded. In cases which involve units of several kinds (both rescue units and ambulances) this list can be rather long.

When the units are sent out, the operator monitors the situation through receiving status reports, either through mobitex messages or through telephone calls. If several operators are involved, i.e. both rescue and ambulance dispatcher, the operators may communicate with each other the status reports that they have got. The unit that reaches the incident first will give a status report and decide together with the operator if the right service has been sent to the accident. All new information is entered into the head of the Resource Form of the case and is thus accessible to the other controllers. The new information may mean that new units are selected and dispatched in the case. It can also mean that some of the units sent are unnecessary.

In the monitoring work, the dispatching operator is working alone, but other operators may verify the status reports from the units. This may be because they want to find status reports aimed for themselves, and just verify the messages that are before in the

incoming line. This does not seem to cause confusion as one might think, since the experienced controllers have a pretty good idea of what the units are doing and how long time it is likely to take. But there are occasions when the status is not updated and a unit is sent out to a new mission without having the status Ready first. When the paramedics then acknowledge a new mission by sending in a status report, an operator not working with them may accidentally put their status as Ready (which makes the new mission be regarded as finished by CoordCom) instead of On The Way and cause some confusion.

The operators' task (among others) is to dispatch and make sure that they have dispatched suitable units. They are not responsible for the actual rescue, just to send proper units. This means that the operators usually do not get informed of the outcome of the actual incident. The case is closed as soon as the units have reported the status Ready (for a new assignment).

Coordinative concerns

The 112 receiving operator (call-taker)

In larger centers, the receiving operator and the dispatcher are almost always two different people. In Stockholm the receiving operator and the dispatcher are never the same person. In Malmö, this could happen provided that the accident has happened in the right area (this is a 1/9 chance, since all operators are allowed to answer 112 calls). While analyzing the case, the operators enter the necessary information in the system. Necessary information is mainly what has happened and where. In order for the operators to decide where an accident has taken place they work with addresses and maps. It is part of the skills to be very familiar with the area. They also try to get a reference point, for example "close to the church" or "in the intersection".

The form that the operator fills in in the CoordCom system orients the questioning quite a lot. The receiving operator knows that in order to do their work, the dispatchers mainly need a priority (1-4), a description of the accident, and a geographical location (including area code). If there is a lengthy driving/parking direction, the operators usually explain it verbally as well. In general, every issue that might be misunderstood or ambiguous is explained verbally, even though there might be a written explanation in the CoordCom case file. It is hard to pinpoint this to a single reason. I interpret this redundant work as a part of the practice in a center of coordination - to *make sure*.

Altogether, the *complexity* of working with receiving 112 calls lies mainly in three issues:

- To collect cues in order to get an intelligible and satisfying description of the incident to base their categorization of the case on.
- To locate the accident.
- To communicate the case to the dispatcher in such a way that the dispatcher has got a sufficient picture of the case.



Figure 11: An operator working with receiving 112 calls

The ambulance dispatcher

The dispatchers' main task is to choose a vehicle, dispatch it, and monitor its development. In order for the dispatcher to be able to send a vehicle, s/he needs to know what has happened and where. Where is more urgent than what in some sense. It is enough to know that it is e.g. a priority 1 (highest priority) and the location. What has happened can be informed while the dispatch is taking place. However, there might be contingencies that make the emergency in need of expertise. One example, described in the Authentic Case chapter, was that a midwife was needed. The ambulances are usually not staffed with a midwife, so s/he had to be picked up.

CoordCom helps picking out an ambulance by listing the vehicles that are connected to the area in question, which might be around 10-15 vehicles. As described by Martin et al. (1997), this is when the real work begins; based on all possible contingencies that are connected to a certain choice, the task is to pick a suitable vehicle. It is very difficult to survey all the aspects that provide the foundation for the choice of a vehicle. I will try to map out some of the ones I have seen in my study here.

- Are the paramedics in a certain unit due for any kind of break soon?
- Is there an ambulance on its way home in this area that we could use?
- Has anyone planned anything for this ambulance?
- Will there be any free ambulances left if I pick this one?
- Is the ambulance adequately equipped?

What I want to point to here is that there is a vast amount of varied kinds of information that the operators might need in order to pick an ambulance. Only a few of them can be drawn from CoordCom or the map computer (if that is in use). The operators also often discuss with their fellow dispatchers. ("Can I take X934?", "is Z943 ready soon?" etc.). The conclusion that can be drawn from the fact that a vast amount of information pieces forms the basis for a decision is that *the varied information has to be easy accessible*. I agree with Martin et

al. (1997) that the map computer as it is provides only a small part of the information that is needed for deciding which unit to send, and it must be faster to locate the information one needs. Perhaps highlighting the ambulances in question could make it quicker to find. For saving time, there should be no time needed to *locate* information, it should be available at a glance.

In this study, the focus is on the relationships between the operators. However, when analyzing their work, it becomes apparent that the coordination between the operators and the paramedics is an important issue. It is my impression that keeping a good relationship to the paramedics is important to the operators. There are some things that might create tension amongst the paramedics, and these are e.g. to get dispatched when they are due to finish for the day, or are supposed to have lunch, and, more importantly, to be wasting time on cases that are not valid. Ambulances are needed for people who cannot walk or sit and need care during the transportation. I have seen cases when the paramedics, if not blankly refuse, but decline to take a patient because s/he can walk. These invalid requests for an ambulance is often the result of a misunderstanding on the part of the emergency caller. Some people also want to go by ambulance because it is free and taxi costs money. (It is hard not to get struck by the impudence of some people when it comes to the use of our common emergency resources.) However, my impression is that sending the paramedics to these invalid cases creates a slight embarrassment for the dispatchers because they (or rather their fellow receiving operator) could not reveal the case to be false.

To maintain a good relationship with the paramedics depends on giving valid assignments. The conclusion drawn from this is that even though the dispatcher can do her work with only the information in the CoordCom system, s/he needs also to be convinced that it is a valid case. The good maintenance of the relationship with the paramedics is based upon the dispatcher believing the case to be valid. Often, the case documentation in CoordCom is enough, but often there is a need for additional information that today is coordinated verbally.

8. Authentic Cases

In the following I present four authentic cases, the first two from the Malmö center and the other two from the Stockholm center. Each of them illustrates different aspects of the operators' work. After each case there is a discussion of issues that are pertinent in that case.

The childbirth case

During this case, I was sitting next to one of the dispatchers who would turn out to be the responsible for the ambulance area in question. Later, I got the opportunity to listen to the recordings of the case. I was sitting next to the operator, co-listening to her conversation through my headset and I was able to see what she was doing on the screen. When the operator got the co-listening request from the receiving operator, it was a man, apparently the husband that was calling in. The dispatcher got the listening in request after the need for an ambulance was established, but since I had the tapes I know how the conversation went on before that.

The wife of the man calling in was one week over time in her pregnancy and the water had broken. One could hear her cry in the background and the receiving operator tried to calm down the situation by telling him that it was a natural process when the baby wanted to come out. Since it was obvious from the beginning that it (if anything) would be an ambulance case, she started to write in an IB-A (resource form for ambulance cases). She entered routinely the data (name, address) that she collected while speaking. Apparently, the husband was rather worried himself and so he asked the operator to talk to his wife directly instead. The operator tried to soothe the woman while the woman told what was going on while crying. When she told that there were large lumps of blood coming out, the operator immediately reacted. She then requested co-listening to the ambulance dispatcher through

the CoordCom system. When the dispatcher answered the co-listening request, she said "I'm with you" as is customary. Otherwise the dispatcher did not speak to the caller but listened only. While the conversation was going on, both the receiving operator and the dispatcher had the case file on their screens.

When the emergency call was finished, the receiving operator entered information into CoordCom and while doing that the two operators discussed through the still open co-listening connection and agreed that they should send both an ambulance and an emergency car (which contains a nurse). It is the dispatchers' job to select the units that would get the assignment. She selected them, and contacted them through radio and sent mobitex messages to them that besides for the Event code also contained the information "large lumps of blood coming out". When these two units were contacted, the emergency car called up and informed, based on the brief information that they had received through radio and mobitex, that the nurse thought that a midwife should be sent to meet them on their way to the hospital. The pregnant woman was living in a small town, Landskrona. They would have to get the midwife from Helsingborg that is about 200 km north of Landskrona, since there was no maternity ward in Landskrona. In order to get the midwife from Helsingborg to Landskrona, the dispatcher needed a fast vehicle, but had no ambulance available. The operator then decided to ask the police in order to get a unit that could transport the midwife from the maternity ward in Helsingborg to the ambulance containing the pregnant woman. The number of the police in Helsingborg was conveniently available in the CoordCom system, so that she only had to search for it and click on the logic number to get the connection. At this stage, it was not yet certain that a midwife was needed, since the ambulance and emergency car not yet had arrived at the woman's house. The receiving police officer promised to have a car ready by the hospital.

While talking to the police, the ambulance had arrived at the scene and found out that the pregnant woman had been seeing a doctor in Lund, which is a city south of Landskrona. The ambulance was therefore taking her to Lund and it was decided that the midwife should also come from Lund and not from Helsingborg in order to meet the ambulance on the way. Another dispatcher (dispatcher2) took this call, so they started to cooperate in order to solve the problem. The first dispatcher then cancelled the police unit and contacted the maternity ward at the hospital in Lund instead of Helsingborg in order to get a midwife that could be sent to the incoming ambulance. While doing that, dispatcher2 contacted an ambulance in the Lund area in order for them to pick up the midwife at the hospital and meet the ambulance coming from Landskrona with the woman. When the maternity ward confirmed that a midwife was available, the dispatcher sent the ambulance to pick her up. While the dispatcher was talking to the paramedics, they picked up the midwife. She asked the dispatcher, on the radio through the paramedics, if this was the woman's first child, which the dispatcher did not know. The dispatcher then gave driving instructions to the paramedics, and then connected the two ambulances' radios so that they could talk to each other. While doing that, she (the dispatcher) also listened to the conversation. The plan was to meet on the motorway and let the midwife go over to the ambulance with the woman.

The emergency car was driving with the ambulance as well. When this switch was done, the dispatcher no longer had to do anything and started to work with other cases.

To me, it was rather stressful not to know how it ended. However, as rarely happens, the dispatcher² actually called when the ambulance that drove the mother reported status Ready (with the assignment). When children are involved, it seems that the operators become more involved than usual. In this case, everything went fine with both mother and baby.

Working with non-professionals demands flexibility

This case is a typical example of the ambiguity in deciding what to do and how to read the incoming information. The talk that is taking place during the emergency call is oriented towards two purposes - one is to fill in the slots that necessarily need information in order for a dispatch to be made. The other is to make the operator content with the case, content enough for her/him to pass it over to a colleague (the dispatcher). This takes some conversation and it is not until the operator starts to talk to the woman that a useful cue is introduced (that lumps of blood were coming out). Further, it was not until the paramedics had arrived and questioned the woman that the information that she was going to Lund instead of Helsingborg became apparent.

We, the citizens, are an unprofessional partner for providing information. We do not, naturally, know how the operators work and what keywords that they need. We are not competent members of the operators' work culture. We do not have a common "language" for accidents. People use for example "it's serious" for very different level of emergencies. The operators need to ask the right questions and find out the conditions that are the foundation for their decisions. This is very much work dependent on experience. The operators sometimes need, like in this case, to talk rather extensively about the conditions in order to get the decisive cue. For medical conditions, they have the medical index that helps them to decide the emergency. This tool is well spoken of, but rarely used.

They also use the CoordCom to categorize the accident. In this case, the operator did not find any post called "lumps of blood" so she had to put pregnancy and an annotation about the special condition in this case. The operators categorize accidents top-down; they have to choose priority in order to be able to categorize the condition in further detail. The system thus groups the information about category and priority. The operators do not equalize priority and type of injury even if the system does so. Other aspects affect the decision for priority, see e.g. The drug abuser case later on in this chapter. In this case, the impression that the couple was very upset probably affected the judgement as well as the mere facts. A more helpful system design would be if the categorization, the typification of the injuries and priority were not connected to each other, because they are often not directly related.

In the case that was analyzed by Whalen et al. (1988), that was presented chapter 6, it was evident that the information or cues needed to satisfy the receiving operator/nurse never came up. Even though the CoordCom forms that support the SOS documentation

work are formal, it is a highly intuitive work. Except from a few "keywords" that generate immediate response (e.g. chest pain, heart failure, no breathing, etc, etc), the operators usually have to make their judgements and decision upon rather varied and uncertain evidence. It is easy to see that the public might be irritated by all these questions, but they are based on the fact that an unprofessionals' assessment is not enough. If one has no medical training, a wound or a situation may appear worse than it really is. Also, some people try to get ambulances although their conditions do not qualify for this for various reasons. The callers sometimes indicate that they get disturbed when getting "questioned" by the operators (like in the Whalen case). This tension suggests that SOS Alarm operator work and the 112 emergency service need to be more visible in society, so that people would know a little bit more about what the service really means. (This might also reduce the number of prank/invalid calls.)

The kitchen fire case

In this case I was sitting next to an operator handling fire rescues. The dispatch in Malmö is done in close collaboration with the fire department (which is located in the same building as the Malmö SOS center, a common arrangement). The white table that can be seen in Figure 2 is the one that the rescue officer on duty uses when there is a larger rescue case. The decisions concerning rescue dispatch always have to be done in close collaboration with the rescue officers. Rescue cases are those handled by what used to be called the fire department, it could be anything from fire extinguishing to rescuing a person that is stranded in a difficult place or even diving after possible drowned persons. Nowadays the fire department in Sweden is named the Rescue Service as a more apt name for what they are doing (and their vehicles hence are named rescue vehicles and not fire trucks).

The call in question immediately caused the operator to make a co-listening request to the operator responsible for rescue cases that I sat next to. It was a rather simple case from the beginning. A woman called in that there was something on the stove that had started to burn and that she could not extinguish the fire. Her son was trying to pour water over it and the operator immediately instructed them to stop pouring water in order to avoid the water to come in contact with electricity. They were instructed to close the door to the kitchen and wait for the rescue service. While talking, the operator had chosen an IB-R (Resource form for Rescue cases) in CoordCom. A detailed driving instruction was written in the form. ("Drive past the big yellow house, turn left, turn right at the large stone on the left side, it is the third house on the left after the church.") She also put down the so called map square number (which people living the country side often know for their house) to pinpoint their location. Based on the information put in the form, the action plan that was generated by CoordCom revealed that three smaller part time rescue stations were to be contacted. Since it was three different stations, there were also several people to inform. The action list that was to be done contained about ten different actions to be made. The operator sitting next to her helped the rescue operator. This was arranged spontaneously.

In order to find the house, the directions were extensively discussed with the different vehicles and units. There was also an ambulance dispatched, as is the rule at a fire in a home. When the first truck arrived, the whole kitchen was on fire, but three trucks were found unnecessary and one was instructed to keep on stand-by instead of driving to the accident by the first officer at the scene. The SOS operator distributed this information to the other units. Since there was no one hurt, it was also suggested that the ambulance could be sent back. The operators role was thus not to make decisions, but to coordinate information between the different parties. She also connected the vehicles' radios so that they could coordinate directly. Her job was to make sure that the right service was there, a little bit like an operating nurse at a distance.

When the fire was extinguished the operator was informed that the rescue team had had to make a hole between the floors to make sure that the fire was out in the roof above the kitchen. An elderly woman was also getting weak and an ambulance was sent for again. The woman did not want to go to the hospital and there was some discussion to and forth on what to do. It was only the practical issues, what connections were made and what resources sent, that were recorded in CoordCom system (besides from the fact that all conversation was recorded). The log of the case thus revealed rather little about the extensive coordination and discussion that the operator participated in.

In order to make sure that the family was all right, the rescue party stayed there and helped the family to contact the insurance company. As the hours passed by, the family got worried of being out of electricity, in particular due to their horse that was about to foal. If there would be no light in the stable, the family would have problems when trying to help the mare. This made the rescue officer contact the operator, asking her to call the closest fire station and request a power generator. The station was unwilling to lend out their power generator, since it was attached to one of their trucks. The operator then contacted the insurance company to ask them for help, which they could not come up with. Finally, the rescue officer and the operator decided that the family would have to manage, and the rescue party left and the case was finished.

Development coordination of a case

Comparing this case with the former, the operator had to take fewer decisions, but had to manage the coordination between the different units. One kind of coordination is the formal status reports that are sent by the mobitex communication system. The other kind is the discussions of the motivations for taking a certain decision. Take for example the ambulance that was first sent back, but due to the rescue officers estimation was called for again since an older lady was getting dizzy. It is notable that the main documentation is taking place in the initial call, after that the status reports are the main part of the event log. The operators have the possibility to document events in free text, a few sentences at a time, but this option is not very frequently used. This means that most reasons for the development are not documented.

This case also illustrates the typical variations and the contextual problems that arise in connection to an accident. The original problem “something is burning on the stove” is by far not the only problem that the operator has to try to get solved. Even if it is not their responsibility, the operators and the rescuers and paramedics seem to reason that peoples' lesser problems have to be solved as well, if there is time. If someone's dog runs away in the middle of an accident, it is likely that the rescue team will try to find it. Besides being a reassuring thing to know, it also points, again, to the rather poor description of the conditions that surrounds an accident. The system should be designed so that it encourages documentation of various kinds so that, in case of staff change, the whole picture is clear.



Figure 12: The operators involved in the drug abuser case (from the Stockholm center)

Since this case got assigned priority 1, the 112 operator made a co-listening (or listening-in) request to the dispatcher, so that when the dispatcher answered to the co-listening request, the case information entered in CoordCom was automatically displayed on her screen. This function can be used while the 112 call is on-going, this was however not the case here, since the 112 operator changed the priority while ending the phone call with the drug addict. Even though the 112 operator can call the dispatcher as an ordinary phone call, she chose to use the co-listening function, as is practice at the Stockholm center. Then the 112 operator knew that the dispatcher got the case information on her screen when they were talking. It was not clear if the dispatcher went into the event log of the case and saw that the 112 operator had changed the priority from 2 to 1. It is possible, but I do not think that she did, she had no reason to do it at that stage. She immediately asked why this case (a “Drug related problem”) got assigned priority 1. In the dispatchers' opinion, from the information she was able to see in the CoordCom case file, the case should have had a lower priority. The 112 operator, however, explained the circumstances that made her choose priority 1 (that she thought the man would walk away if not picked up immediately). With this information, they finished the co-listening and the dispatcher started to dispatch an ambulance.

Accountability of decisions

When a piece of information is entered in the system it has a special meaning to the operator who entered it. The information is then travelling to another context, since the context is changed, which may give it a slightly different meaning, so that the case file can in a sense be understood as a boundary object. Thus, the context is an important part of interpreting the information. In the case presented above we have three conversations, three coordinations of information. Firstly, there is a conversation between the caller and the 112 receiver. Secondly, there is a computerized communication from the 112 operator to the dispatcher. Thirdly there is a verbal communication between the 112 operator and the dispatcher. These three have different contexts that are intertwined with the choices and explanations. Drawing on the content of the last conversation between the operators, the conversation between the caller and the 112 receiver documented in the computer system misses some context for the dispatcher to make sense of it. She needs some contextual clues from the conversation to understand the decisions that came out of it.

From what this example shows, the decision-making often needs to be verbally supported and coordinated amongst the controllers. It is my impression that the belief at SOS Alarm, both amongst the controllers and the management, is that the *system* supports all the information that needs to be coordinated. However, this is not the way they work, at least not with priority 1 cases. The information that is transferred during the co-listening phase often adds pieces of information to help the dispatcher to make sense of the case that they would not have access to if it were not for the verbal exchange.

The operator has the accident priority and categorization in the CoordCom to go by but it seems that this is mostly used for describing the accident and not used as a decision

tool. To decide what kind of priority that will be assigned to the case is entirely based on their impression of the emergency call/caller and the SOS practice as with The childbirth case above. The receiving operator has some rules of thumb to go by. These are not so easy to grasp, since each case has to be evaluated uniquely. One basic rule of thumb is to make sure if an ambulance is needed. There is only a limited number of ambulance units available so they have to be distributed with care. Another rule of thumb is that accidents in public places get higher priority, because they create more commotion and are more awkward for the injured person. These issues are what possibly can be understood as general motivation for the first and second decisions in the drug abuser case. When it comes to the third decision to change the priority to 1, it is more difficult. The operator gave me an explanation that caller is likely to disappear, but how did she draw that conclusion? It might be the speeded but polite voice that made the call. It might be that I was sitting next to her, and she wanted to be sure not to make any mistakes. It might be that she felt sorry for the man and wanted to help him. (I believe compassion to be one of the main parts affecting the decision-making, e.g. cases with children are handled with thorough resources). It might be that she is rather new at SOS.

When the dispatcher got the case on her screen through the listening in function, she asked why this case had gotten priority 1. I find it interesting, when analyzing the interface, that there has been a belief that the uniqueness of the accident is supposed to be explained by three numbers and about 30 letters of free text. Perhaps this generalization of incidents is necessary for making the cases manageable. On the other hand, the interface and documentation possibilities do not always (like in this case) support the common understanding of what is being done. It may also be difficult to get back to a certain case and understand how the operator was thinking (although they do have tape recordings of all conversation).

In conclusion; this case shows two things:

- The circumstances of the problem are as important as the conditions of the injuries when the operators decide priority.
- The operators do coordinate issues verbally that are not entered into the CoordCom.

If the verbal coordination is to be reduced then the cases need more documentation of circumstances taken into account.

The fever child case



Figure 13: The 112 operator, physician and dispatcher involved in the fever child case

The operator I sat next to was working as a 112 receiver, that is he only received 112 calls and passed them on to the dispatchers. He was working in the position at the lower left dot in Figure 13. The woman who called 112 presented herself as the nurse in the medical rescue boat that is stationed in the archipelago of Stockholm. She was with a physician who during the conversation took care of a little boy of 10 years old. The boy had high fever and the physician thought that there was some kind of stomach problem that was named in Latin. The question that the nurse put forward was how emergent this patient should be treated and how should he get transported to the Astrid Lindgren children's hospital. The 112 operator was not that familiar with the stomach problem the nurse mentioned, so he did a co-listening request to the physician in the opposite part of the room (the topmost in Figure 13).

The physicians work at the Stockholm center at daytime, but while they are using the CoordCom to follow the cases and support the operators with decisions, they do not have any formal education on the CoordCom system. This meant that the 112 operator had to follow the conversation and enter the decisions that the physician took. In order for the physician to help the operator collect the right information, he walked from his position to the 112 operator. The physician was using a cordless ordinary phone (as opposed to the ordinary headset). The nurse at the other end had turned over the phone to the archipelago physician at the sick boy's house. The two physicians then discussed over the phone while the SOS physician stood next to the 112 operator (who was still co-listening). The question was; is the boy fit enough to take the ambulance boat or should they send the rescue helicopter? They decided that while the boy might be fit to go by the boat (which took longer time) he should not have to suffer the rather bumpy ride over the waves. The helicopter would pick him up (and thus the case got priority 1). After this decision was made, the 112 operator continued to talk to the nurse that collected information from the boy's parents on where the rescue helicopter could land. When the call was finished, the 112 operator did a co-listening request to the dispatcher (who dispatched not only ambulances on wheels, but boats and the helicopter as well). While talking on the intercom (through CoordCom), where

the receiving operator explained the specifics of the case, she suddenly unplugged her headset and walked over to the dispatcher to explain the helicopter landing details. It was not a map that needed to be coordinated, which might have explained the receiving operators' need to walk over to the dispatcher. Instead, my interpretation was that the receiving operator wanted to give a longer explanation and she did not want to occupy the dispatcher's headset with their conversation. When talking outside the headsets, the dispatcher was able to communicate with the 112 operator while dispatching the helicopter, using the system.

The importance of local knowledge

Each center has special knowledge of the conditions in their area. In this case, the possibility to dispatch a helicopter was used. For the Stockholm area with its vast archipelago this is an important unit. The local variations more or less taken for granted are extremely difficult to pinpoint, and to remember that others may not know. Considering a future system where calls and cases may be handled from other centers, this seems to be an important issue. What does these local differences mean for SOS centers and what kinds of differences are there? Local knowledge can mean many things but generally aims at information that is physically limited to a certain geographical location. I have divided this discussion into three areas; geographical, cultural and temporary local knowledge.

The main geographical issue is to *know the area*; the main roads and their names, where the emergency hospitals are, the names of different areas and so on. This seems to be especially important for the Receiving operators in Stockholm. They always ask for the name of the area in as much detail as possible, e.g:

Op "U-hu, so it is in Jakobsberg, is that in the Center, or..?"

C1 "Yes, it is close to the /train-/station."

Op "Close to the station, okey."

Informal names and places is another issue:

An operator in Växjö received an overflowing 112-call from Malmö (currently, phone calls may be received at another center but they can do nothing about it but to ship it back to the center in which it belongs). The man calling said that his friend had fainted, fallen down (priority 1) "in a park here in Malmö". The operator asked which park. The man answered:

"You know, the one where one plays petanque in here in Malmö." The operator did not know where this was but had to rely on that this information would make sense to dispatchers and paramedics in Malmö.

This kind of local knowledge, "the park where we play petanque" in a city, is of course a very difficult type of information to spread. During one discussion an operator suggested that local information like this could be added to the map computer and of course it could, but it would probably take a lot of time. My belief is that this kind of information needs to be coordinated verbally between centers. It may be included in the role of the shift leader to be available as a resource of geographical local knowledge, perhaps through a chat system.

Another local knowledge issue is to be able to understand (and perhaps communicate in) the local dialect. Some of the operators in Malmö have told us that they are worried about misunderstandings that can occur due to different dialects e.g. between the north and the south of Sweden. Already there is an issue when dealing with different foreign accents and people who need a translator. If this were to be a common phenomenon, that the information exchange were problematic to understand, it might lead to a larger workload and potentially more miscommunications.

There are several aspects that are associated with the local, common room, the culture at a certain workplace. As described in the studies above, there are large differences of the practices in between the different centers, this is what I call cultural local knowledge. The education of the operators is another issue, how tasks and artifacts they know differ. The operators also claim that knowing one's colleagues, knowing how the fellow operators "think, sit and do", their style, is essential for being able to coordinate the work. Another kind of cultural local knowledge is to know repetitive callers and not to send out any rescue to them (some people call 15 times a day). It is also cultural knowledge to know something about the units, their typical schedules and the paramedics.

Typical local knowledge in connection with time is to know when the city is clogged with commuters, which may need different judgements of which ambulance to pick. Other typical knowledge of temporary circumstances is to know if there is road work being done that may hinder the traffic, or if there is a large event going on, and a lot of people in a certain area, e.g. at a larger concert or sport event. Of course, living in the same city as one is working in helps the operators in their work. The information about what is up is not only relevant to them at work, but also as private citizens. I have noticed that living in a certain area, say e.g. Södertälje, makes the operator a bit responsible for knowing the area. Knowing where your colleagues live, or are especially knowledgeable about is certainly another kind of local knowledge.

In total, there are at least three different kinds of knowledge that is connected to the local center but not accessible in any proceedings or documentation. These are

- knowing one's area; geographical specialties like being next to the sea, or mountains, or having extreme weather conditions like in the north and how to handle emergencies in connection to the local situation; and to know the district; well known spots, names, roads, the location of hospitals, etc.
- understanding the local context; cultural differences between centers; dialects; knowing the style of others; using the same tools
- knowing what is going on currently in the area one works with, large festivals, road constructions, commuting ours, etc

How can these differences be bridged in a future system without creating misunderstandings? One thing, as I mentioned, is to make one operator, or the shift leader, a contact person for information. But it is hard to look for information that one does not know exists. How could an operator from a center in the middle of the country be aware of the possibility to use a helicopter, as in the example above? There are many problems to be solved here; the operators must not be overwhelmed with information because that will slow

them down. One thing that is sure from this study is that the case files must be clearer so that the fellow operator can easily see the background to the decision made by the receiving operator and be content that it is valid.

9. Designing technology for centers of coordination: preliminary considerations

This chapter describes characteristics that are typical for work in centers of coordination, (control rooms) and touches briefly on the consequences of them for the design. There are certainly other aspects as well, but my studies, both of the SOS Alarm and the earlier of air traffic control, points towards four aspects as central in forming the possibilities and constraints for design in these environments. These are:

- Time critical tasks (having absolute deadlines)
- Distributed setting (the effects of/problems with distributed collaboration)
- The complexity and degree of unpredictability in the work
- Extensive real-time information exchange (environments with elaborate need for updates).

The typical setting for a controller/operator in command and control centers includes arrays of information sources and communication tools. Usually the operators have "positions" where the specific tools are disposed so that anyone can easily take over the position with a minimum of adjustments.

Two typical control room positions:



Figure 14: An example of a non-computerized setting with a vast array of tools

Figure 15: A computerized setting at SOS Alarm (still with some tools on the side).



Even though the CoordCom system that the SOS operators use has an advanced design, the sources outside the system are not totally removed as you can see by the paper sources lying around at the desk in Figure 15. Overview and easy access seem to be the reason for maintaining sources outside of the computer system.

In the following, I will discuss some characteristics of the work in relation to the characteristics of the tools.

Time-critical work

Time-critical work is work that concerns an ongoing process that the actor cannot stop, for example air traffic controllers that manage the flights. If they would leave their positions for a while the flights would continue as opposed to for example a designer that can leave sketches on the desk that will not change until one gets back. This means that time-critical work is paced by a more or less uncontrollable factor (the airplanes in the sky or the incoming accidents at SOS Alarm, for example). A professional that does time-critical work needs to have all the information when it is needed, they cannot wait until someone else is finished.

In order to be in control, the actors need to know what is happening at the same time as it is happening. The information also needs to be easy accessible. Simple graphics are preferred before advanced representations, see Perby (1995). Text representations or even DOS-interfaces with no graphics at all are often preferred since the information is easy to find. Professionals doing time-critical work tend to want to boil down the information to the smallest possible form as we have seen in the SOS study as well as in the ATC study.

The possibility of finding information fast, *at a glance*, is essential. My impression from the studies of air traffic control and SOS Alarm is that a typical stress source is when information has to be searched for.

Besides the visual output, time-critical work also needs to be considered when designing for the possibilities of input. In air traffic control, the tangibility of the flight strips has been discussed (e.g. Berndtsson & Normark, 1999; Shapiro, Hughes, Randall, & Harper, 1993). However computerizable or not, it is clear from the studies that the easily accessed paper flight strips, among other things annotated with updates of the orders given to the pilot, have a design that make them very useful for their purpose. One source of delay seems to be when shifting from one input device to another, for example when changing from a pen to a keyboard or from a mouse to a keyboard. On the other hand, extensive use of the keyboard for moving around in the interface by for example using the Tab key as the SOS operators do seems to be a bit monotonous.

I believe that interaction with tools is largely supported by the involvement of our bodies. If one studies the interaction between the air traffic controllers and their tools, it is easy to see that they have developed series of bodily routines of looking at the radar, turning to the flight strips, turning to other informative screens, and so on. I believe that they involve their bodies when juxtaposing information, the movements are a part of their selection process. They do not have to *think* of where to look for a certain piece of information, they can to some extent trust their bodies to know. This aspect creates a challenge for computerized designs that currently, like at SOS Alarm, present all information on one screen. I think that the varied setting of the old control rooms, see Figure 14, is much better from this point of view than the newer ones, however more effective in other ways (for example in information storing).

Distributed work

When using the term distributed work, I mean a work setting where the actors must interact with each other by shared tools and through telephone. The operators at SOS Alarm cannot normally walk over to each other if they want to talk about something so they have to depend on tools. Distributed work demands shared information spaces and/or tools that support verbal or textual exchange. Often, control room work is distributed in the sense that the actors are not allowed to talk out in the air but have to use the systems in order for the conversation to be recorded.

Since control room work is characterized by a strong integration (or coordination) of work between the actors, the need to know what others are doing is great. The work done by one actor is the foundation for the work of another. Therefore, the tools that distribute what an actor does, need to be clear and unambiguous. Preferably others should be able to, based on the information displayed, derive how any given colleague was thinking while doing his/her work. This is something that I will discuss further in the SOS study, but mainly, the acceptance of other's decisions lie in the knowledge of the issues that the decisions were based on.

The updates of others' work need to be instant. In air traffic control, they have decided not to give any intrusive indication through the tools that something has been done. There is no beep or alarm going off. Instead they use a camera system that distributes the state of the annotated flight strips as they are managed. It might be that the level of prediction is so high so that the controllers do not need any obtrusive indication that something is happening. They also have other sources that tell them about what will happen in the next couple of minutes.

The controllers talked a lot about interruption in the work. They wanted to reduce all unnecessary interruption that took place when coordinating work that was within the "normal case", what was to be expected. The same is taking place at SOS Alarm in Stockholm. All cases that are not of highest priority are just transferred through the system from the receiving operator to the responsible dispatcher. S/he sits looking at the current cases list and acts when there is a case labeled the right priority for her/him, without any special notification from the operator that received the emergency call.

For distributed work at control rooms to work, the means of distribution seems to be relying on a "normal situation", when it is unambiguous and "as it should be". It is only when there is ambiguity, that the information needs to be explained with additional verbal exchange. The actors do, to a large extent, trust the tools to distribute the information to their colleagues.

Unpredictable work

Control room work is work of maintaining control over something that often becomes, if not dangerous, at least destructive if the control is lost. For SOS operators the thing to maintain control over is, generally speaking, the citizens safety; the rescue in accidents and

incidents in their area. For air traffic controllers, it is safe non-conflicting routes for the airplanes in their area. The complexity of the work is severe, because of the infinite circumstances that might occur, at a moment's notice. In order to reduce the complexity, the actors must have reliable regulations and tools that inform and help them to predict the future situation.

Because both the character and load of work is unpredictable, the actors need to keep an edge, a preparedness, for a situation when it comes. It is somewhat "dangerous" with tools that warn too much about possible conflicts. The air traffic controllers, for example, get a warning when two airplanes are on conflicting routes on the radar. This warning is based on the airplanes' current heading and takes no notice of the fact that the controller already had planned them to get new directions in a moment. The alarm thus goes off even though the situation is already considered. The SOS operators get a lot of so-called silent calls and prank calls that are of course not due to technology, but, according to the operators, make them less alert and their preparedness for extreme situations lower.

It is rather important for the professionals to have access to the facts that their colleague based their decision on. In every deviant case, the attentive actor is likely to ask why. If the main reasons for why a certain decision was made were to be displayed somehow in the artifacts, I believe that this would reduce the need for verbal communication. The actors say that it is somewhat time-consuming, and as have been seen in this study, it occupies the headset so that conversation with others have to wait. Considering the SOS Alarm setting, I believe a step towards a less injury-oriented interface design and more orientation towards the whole situation, should be tried out.

It is important for the technology and tools in control rooms to help the actors to maintain their sensitivity towards the information that they receive, but at the same time the information needs to be unambiguous. It is the professionals that should evaluate the situation. If the technology does it for them, there is a potential risk that the actors will not notice a problem if it is not spelled out to them.

Real-time information exchange

This aspect has been a part of all the issues above. On a more concrete level, the different sources for communicating are basically of these categories; shared environment, shared information spaces, written and verbal exchange. As many researchers have noticed, e.g. Harper & Hughes (1993); Heath & Luff (1991a); Norman (1993), the common environment - the possibility to see what others are doing - is fundamental for coordinating work. The movements of others are notifications that something is happening. Shared information spaces, for example the current cases list in the system at SOS Alarm, are useful for getting an awareness of the activities of others. In control rooms, the things that need to be coordinated verbally, that are out of the ordinary, are of great interest. Text chat is too time-consuming, instead actors mostly talk on some kind of intercom system. One thing is clear from my studies; no setting can rely on computerized information exchange only. There is a need to *speak* to the colleagues in certain cases and there always will be.

Towards computer supported control room work

What can be seen for example in the SOS study and in future ideas for e.g. air traffic control is the interest to bring every tool and information source into a computerized system. The systems are tempting from the perspective of effective and fast information sharing, retrieving and storing. However, when working with one screen, all information cannot possibly be shown at once. Even if the screen is large enough to show several small juxtaposed windows with different pieces of information, *it demands a more detailed and active looking* than the casual, at a glance, view of a certain tool that the former settings provided. The strong affordances of non-technical tools like a flight strip and the possibilities for bodily orientation towards the pieces of interest are useful both for coordination and orientation.

Of the four characteristics that I have discussed here, I believe the time-criticalness to be of greatest influence. Every problem can be solved if one gets sufficient time to do it. The professionals in these settings are working with an uncontrollable situation where the workload and kinds of problems always vary. Being able to reduce the tasks to a minimum is important in these settings. However, one should not forget the educational possibilities that are created when discussing a situation. I earlier discussed the varied nature of local knowledge at a SOS center. This is created by the discussion about cases.

10. Concluding discussion and future work

The newer computerized settings changed many of the procedures that took a long time to develop. It altered the very foundation for many professions. At this stage we have learned that computers in general and digital solutions in particular are here to stay. The current setting, with a keyboard and mouse as the main input devices, and one screen as the main output device, is in my opinion the first shaky steps toward digitized tools. The potential of digital augmentation of tools is yet to be seen. In the meantime we need to understand how computer systems are designed and used in the everyday setting, for routine everyday work. As we have seen in the study of SOS Alarm, the system interface is much based on former non-computerized fill-in forms. Often, the operators work with a small part of the screen, the case is so to speak moved around on the screen. This is what the operators work with, is it sufficient for their needs?

The thesis' main parts were the theoretical background and the SOS Alarm field study. In the theoretical part, the main concerns were workplace studies and collection of issues from other studies as reference points for ethnographic analysis. Due to the discussion about what ethnography can contribute with, it has been suggested that relevant concepts for guidance may be one way to augment the use of ethnographies for design (Bannon, 2000). I agree with this idea and gave some suggestions; tool-mediated coordination, coordination mechanisms, awareness of the activities of others, juxtaposing of information sources and boundary objects as some interesting issues that have been discussed earlier.

One general conclusion that can be drawn from the field study is that the system handles facts well. Facts are, however, not independent of the context. Impressions, values

and local knowledge are important and unconditionally a part of the SOS Alarm operators' work. The computerized system, as it looks today and what seems to be realistic with current technology, should be considered as a communication and information storing system. It should not take any responsibility for categorization and analysis of the case. One conclusion drawn from this is that the case files, as they look today, are not complete. They lack important information, conditions that are not considered as facts and thus not entered into the system. There are some cases when the information in the system is sufficient, but in many of the cases, the operators have to coordinate verbally. There have been no cases in the study where the verbal coordination has been a problem, but, when considering a new setting where temporary coordination can be arranged at distances, there are many. One of the main potential problems is the time constraint. When the operators need to coordinate verbally they can smooth the flow by seeing what the other one is doing. They have overview of the current situation at the center and the actions of others. This will be more difficult, but not unsolvable, in a distance coordination arrangement. Nevertheless, verbal coordination is important for common understanding. The goal should not be to remove it, but to consider when it can be replaced by a more nuanced case file information.

The account of the work of the receiving operator and the dispatcher and the CoordCom system gave several different results. The system is founded upon two older tools, namely the old switchboard and a paper form. The interface is thus not effectively used, the operators work with only a part of the screen so that the full potential of the interface is not used. The form-oriented documentation part (see e.g. Figure 8) calls for keeping the information short, too short for some occasions. The system's categorization method is based on a relationship between the priority of the case and the type of injury that the person in need has. This is often, as I have tried to show, worked around since it is not only physical facts but conditions of the caller (is s/he for example vague or upset), the context of the accident (is it a public place), the age of the injured, and many other reasons that decide the priority. Also, in rescue cases, it is only the facts and not the underlying reasons that are documented. If this design is maintained, I see a potential problem when cases are supposed to be coordinated between centers and this important information will be lacking.

The receiving operator documents facts, such as the address of the incident and the name of the injured and so on. That is the easiest part of this work. The more complex work lies in interpreting the caller's information and to collect sufficient information so that the call can be accepted or rejected as a case. It is not only that the receiving operator needs to be sure, s/he also needs have convincing arguments for the dispatcher when the case is handed over.

Further, the dispatcher on the other hand needs to be sure that the case is valid so that the ambulance is not turned out on an invalid case. The process of selecting which unit to send is guided by many properties of the units, of which the location is only one.

The case analysis gave four different aspects of the operator's work of which the last two were most important; the accountability of decisions and the importance of local knowledge. Today, if the account for a certain decision does not lie in the most general facts

in a case (for example the kind of injury), the receiving operator and the dispatcher coordinate verbally. The dispatcher needs to make sense of a decision in order to be content in dispatching a unit. The local knowledge is another aspect that is important for the operators in order for them to efficiently handle a case. There are at least three kinds of local knowledge; geographical, temporary and cultural, that are of importance for the coordination of work.

Control rooms are sometimes referred to as centers of coordination. The expression is useful to describe the interest I take in these sites; they are intensively coordinative and do exhibit a need for interpersonal awareness and real-time interaction. In my studies, both the one presented here and the earlier on air traffic control, I have come to the conclusion that four aspects at centers of coordination, closely related to each other, are useful for describing the work that is going on there. These are

- working with severe time constraints
- distributed work
- complexity because of the unpredictability
- need of real-time information exchange

The interaction with technology in centers of coordination is in my opinion very much formed by these four criteria. Problems that occur in the work can often be derived to any of these criteria. The more intensely they are considered in the design of tools, the smoother the work seems to get.

The SOS Alarm has been a most interesting and very complex setting to study. I feel that I have learned many things, not only about the setting but also about the possibilities and constraints of my data collection techniques. On a personal level it has also been interesting to study how this public service works in practice.

For the future, I would like to find ways to approach design problems in more detail. I have touched upon the interface design of the CoordCom and some functionality problems and limitations in this thesis, but I would like to take design discussions a step further, perhaps towards new design suggestions. This means to put on different hats so to speak and work with the two roles of the ethnographer and the designer in parallel.

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